

Distributed Entropy-Regularized Multi-Agent Reinforcement Learning with Policy Consensus

Yifan Hu ^a, Junjie Fu ^a, Guanghui Wen ^a, Yuezuo Lv ^b, Wei Ren ^c

^a*School of Mathematics, Southeast University, Nanjing 210096, China.*

^b*Advanced Research Institute of Multidisciplinary Sciences, Beijing Institute of Technology, Beijing 100081, China.*

^c*Department of Electrical and Computer Engineering, University of California, Riverside 92521, USA.*

Abstract

Sample efficiency is a limiting factor for existing distributed multi-agent reinforcement learning (MARL) algorithms over networked multi-agent systems. In this paper, the sample efficiency problem is tackled by formally incorporating the entropy regularization into the distributed MARL algorithm design. Firstly, a new entropy-regularized MARL problem is formulated under the model of networked multi-agent Markov decision processes with observation-based policies and homogeneous agents, where the policy parameter sharing among the agents provably preserves the optimality. Secondly, an on-policy distributed actor-critic algorithm is proposed, where each agent shares its parameters of both the critic and actor for consensus update. Then, the convergence analysis of the proposed algorithm is provided based on the stochastic approximation theory under the assumption of linear function approximation of the critic. Furthermore, a practical off-policy version of the proposed algorithm is developed which possesses scalability, data efficiency and learning stability. Finally, the proposed distributed algorithm is compared against the solid baselines including two classic centralized training algorithms in the multi-agent particle environment, whose learning performance is empirically demonstrated through extensive simulation experiments.

Key words: distributed actor-critic algorithm, networked multi-agent system, entropy regularization, deep reinforcement learning

1 Introduction

Reinforcement learning (RL) has emerged as a powerful technique for enabling artificial intelligence to learn from experience and optimize decision-making in various complex, dynamic environments, which has achieved dramatic performance in a variety of fields, such as optimal control [1], robotics [2] and cyber-physical systems [3]. These advances have sparked widespread interest in applying RL to multi-agent systems. In particular, multi-agent RL (MARL) has shown tremendous promise in enabling groups of agents to learn effective coordination strategies [4]. However, it remains a significant challenge to design sample efficient MARL algorithms for multi-agent systems in practical physical environments.

Inspired by the impressive advantages of scalability and robustness of distributed coordination [5–8], the recent research of MARL has focused on the design of algorithms which can learn decentralized policies directly. Considering the non-stationarity of the environment due to the policy changing of agents during the training process, the paradigm of centralized training with decentralized execution (CTDE) has attracted increasing attention, which eliminates this issue by employing a centralized trainer to learn a global (action-)value function. With the combination of the deep RL (DRL) training architecture, various efficient algorithms have been proposed under this paradigm [9–11]. It is worth pointing out that the observation-based policies are widely considered in these algorithms, where the agents are able to behave differently using the same policy under a common state. This motivates the utilization of the policy parameter sharing mechanism in the algorithm design [12, 13], which significantly improves the learning efficiency as well as the scalability of the MARL algorithms. However, due to the centralized learning architecture, these CTDE-based approaches will be imprac-

* This paper was not presented at any IFAC meeting.

Email addresses: yfhu@seu.edu.cn (Yifan Hu), fujunjie@seu.edu.cn (Junjie Fu), ghwen@seu.edu.cn (Guanghui Wen), yzlv@bit.edu.cn (Yuezuo Lv), ren@ece.ucr.edu (Wei Ren).

tical when no centralized unit exists to perfectly process the information of all agents.

It has drawn increasing attention to design distributed RL algorithms over networked multi-agent systems over the past decade [14, 15]. A class of distributed algorithms has been proposed to solve the classic single-agent RL problem [16–18] and its variant multitask RL problem [19–21]. However, these distributed algorithms may not be able to tackle the MARL problem which the CTDE-based approaches concern due to that each agent here handles an independent Markov decision process (MDP) without considering the influence of other agents. To enable distributed policy learning in the environment containing multiple decision-makers, another class of distributed algorithms was designed based on the MARL model networked multi-agent MDP (N-MAMDP) [22–26]. Compared with the CTDE-based algorithms, the distributed algorithms allow each agent to update its own policy to collaboratively achieve the team goal only with the locally collected data and the information from its neighbors, which have shown the real-world application potential in the fields such as multi-robot coordination [26] and wireless communication [27], where the practical constraints like privacy protection and [limited](#) communication bandwidth should be considered during training. In [22], each agent obtains a local approximator of the optimal joint policy, which is updated based on the gradient temporal difference (TD) learning and the parameter consensus. In [23], an on-policy distributed actor-critic algorithm was developed, where the consensus update step is applied to the critic parameter such that each agent is able to approximate the global action-value function using local information. Furthermore, this algorithm was extended in both the off-policy setting [24] and the continuous action space setting [25]. Compared with the CTDE-based MARL algorithms, these distributed MARL algorithms [22–25] enjoy the rigorous theoretical convergence guarantee under the linear critic assumption, but their learning performance is commonly validated only on toy environments. Recently, the distributed MARL problem was further studied in the N-MAMDP with homogeneous agents and observation-based policies [26], where the policy parameter sharing mechanism considered in CTDE-based approaches was proven to preserve the optimality of the objective. Based on this observation, a novel distributed algorithm was developed in [26] by incorporating the policy parameter consensus step into the algorithm from [23], which is also convergent. By employing the DRL architecture, it is shown in [26] that the proposed algorithm has comparable performance to the vanilla CTDE-based baselines on standard MARL tasks.

Sample efficiency plays a critical role in RL, where the ability to learn from a limited number of samples is essential for RL algorithms to scale up and tackle complex problems efficiently. Over the past few years, the entropy regularization has been verified as a powerful technique

to improve sample efficiency in RL, which incentivizes the policy to explore the environment widely with sufficiently random actions by adding a penalty term based on the policy’s entropy to the RL objective. Based on this technique, a variety of practical algorithms have been proposed including single-agent RL algorithms [28, 29] and CTDE-based MARL algorithms [30, 31]. Apart from its efficient applications, the theoretical foundations of single-agent entropy-regularized RL have also been established [32, 33]. However, entropy regularization has not been incorporated into the distributed MARL algorithms so far due to both the algorithmic and analytical challenges. Inspired by the policy gradient theorem and the provable actor-critic algorithm proposed for the single agent case in [33], we consider using entropy regularization in the design of new practical distributed MARL algorithms.

In light of the discussion above, we investigate the combination of the entropy regularization RL architecture and the distributed MARL algorithm. A distributed entropy-regularized actor-critic algorithm is proposed containing the consensus update of both the critic and actor, which obtains the convergence guarantee under standard assumptions. Compared with the existing results, the main contributions of our work are summarized as follows:

- (1) An entropy-regularized MARL problem is formulated, where the parameter sharing policy is proven to preserve the optimality of the objective for homogeneous multi-agent systems. Compared with [23, 26], the more challenging task of learning policies which jointly optimize the average return and the entropy of the joint policy in a distributed manner is considered.
- (2) A local entropy-regularized policy gradient is proposed which extends the results in [33] from the single-agent RL setting to the MARL setting. Based on the proposed policy gradient, a convergent on-policy distributed actor-critic algorithm is developed, where the entropy of the joint policy is estimated by each agent with its own policy for local actor update, and the consensus update is applied to both the critic and the actor.
- (3) Compared with [26], the sample efficiency issue is specifically tackled with the entropy regularization technique. Furthermore, the convergence of the local actor parameters is analyzed in more depth from both the theoretical and empirical perspectives.
- (4) A practical version of the proposed algorithm is designed based on the DRL training architecture, which is also distributed and enjoys the advantages of scalability, data efficiency and learning stability. The experimental results demonstrate that the proposed algorithm has comparable learning performance to the state-of-the-art CTDE-based algorithms on standard cooperative MARL tasks while only requiring distributed interaction.

The remainder of the paper is constructed as follows. In Section 2, some preliminaries are provided. In Section 3, the entropy-regularized MARL problem is formulated. In Section 4, the distributed actor-critic algorithm is proposed. In Section 5, the convergence results for the proposed algorithm are provided. A practical distributed algorithm is developed in Section 6. Experimental results are given in Section 7. Finally, concluding remarks are provided in Section 8.

Notations: For a finite set $\mathcal{Z}$, its cardinality is represented by $|\mathcal{Z}|$. Denote the sets of real numbers, n -dimensional real vectors and real matrices with n rows and m columns by $\mathbb{R}$, $\mathbb{R}^n$ and $\mathbb{R}^{n \times m}$, respectively. For a vector $x = [x_1, \dots, x_n]^T \in \mathbb{R}^n$, $\|x\|$ represents the Euclidean norm. Denote $\mathbf{1}$ and I as the all-one vector and the identity matrix with proper dimensions, respectively. Denote $\mathbb{I}_{\{\cdot\}}$ as the indicator function, $\mathcal{N} = \{1, \dots, N\}$, and $M = [m_1, \dots, m_N]$ as a permutation, which satisfies $m_i \in \mathcal{N}$ and $m_i \neq m_j$ if $i \neq j \forall i, j \in \mathcal{N}$. Then, a permutation of any ordered list $y = (y_1, \dots, y_N)$ under M is denoted by $My = (y_{m_1}, \dots, y_{m_N})$. Denote $\mathcal{M}$ as the set of all possible permutations.

2 Preliminaries

2.1 Graph Theory

The interaction among the N agents can be modeled as a time-varying undirected graph $\mathcal{G}_t = (\mathcal{N}, \mathcal{E}_t)$, where $\mathcal{N}$ denotes the vertex set and $\mathcal{E}_t \subseteq \{(i, j) : i, j \in \mathcal{N}\}$ denotes the edge set. Each agent i is represented by a vertex i . Agents i and j can share information with each other at time t if $(i, j) \in \mathcal{E}_t$. Denote $\mathcal{N}_t^i = \{j : (i, j) \in \mathcal{E}_t\}$ as the neighboring set of agent i at time t . Given a graph $\mathcal{G}_t$, we can define a weight matrix $C_t = [c_t(i, j)]_{N \times N}$, where $c_t(i, j) \geq 0$ for any $i, j \in \mathcal{N}$, $c_t(i, j) = 0$ if $(i, j) \notin \mathcal{E}_t$. C_t is said to be row stochastic if $\sum_{j \in \mathcal{N}} c_t(i, j) = 1$ for any $i \in \mathcal{N}$ and column stochastic if $\sum_{i \in \mathcal{N}} c_t(i, j) = 1$ for any $j \in \mathcal{N}$.

2.2 Networked Multi-Agent Markov Decision Process

Consider an environment containing N agents with a time-varying sparse communication network $\mathcal{G}_t = (\mathcal{N}, \mathcal{E}_t)$. We can build the cooperative MARL problem on the model N-MAMDP [23] which is defined by a tuple $(\mathcal{S}, \{\mathcal{A}^i\}_{i \in \mathcal{N}}, P, \{R^i\}_{i \in \mathcal{N}}, \gamma, \{\mathcal{G}_t\}_{t \geq 0})$, where $\mathcal{S}$ is a finite global state space, $\mathcal{A}^i$ is a finite action space of agent i , and $\gamma \in [0, 1)$ is a discount factor. Let $\mathcal{A} = \prod_{i=1}^N \mathcal{A}^i$ be the joint action space of all agents. Then $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ denotes a state transition probability function, and $R^i : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ denotes a reward function of agent i . Denote the local policy of agent i by $\pi^i : \mathcal{S} \times \mathcal{A}^i \rightarrow [0, 1]$. At each timestep t , each agent i samples an action $a_t^i \in \mathcal{A}^i$ from the distribution $\pi^i(\cdot | s_t)$, where $s_t \in \mathcal{S}$ denotes the global state at

that moment. After the joint action $a_t = (a_t^1, \dots, a_t^N)$ is executed, the N-MAMDP will shift to a new global state $s_{t+1} \in \mathcal{S}$, and each agent i will receive a reward r_{t+1}^i , which satisfies $R^i(s_t, a_t) = \mathbb{E}[r_{t+1}^i | s_t, a_t]$ for any $i \in \mathcal{N}$. It is assumed that each agent can observe the global state and the joint action, but it has no access to the rewards and the policies of the other agents. In addition, the agent can share its local information with its neighbors through the communication network $\mathcal{G}_t$.

3 Problem Formulation

In this paper, we consider N homogeneous agents interacting with the environment to learn collaborative policies. Due to the homogeneity of the agents, the MARL problem can be modeled as a subclass of N-MAMDPs, named homogeneous N-MAMDP [26], which is defined as follows:

Definition 1 A networked multi-agent Markov decision process $(\mathcal{S}, \{\mathcal{A}^i\}_{i \in \mathcal{N}}, P, \{R^i\}_{i \in \mathcal{N}}, \gamma, \{\mathcal{G}_t\}_{t \geq 0})$ is homogeneous if

- (i) Each agent has the same local action space, i.e., $\mathcal{A}^i = \mathcal{A}^j \forall i, j \in \mathcal{N}$. Furthermore, the global state space $\mathcal{S}$ can be decomposed into N homogeneous local state spaces $\mathcal{S}^i$ such that $s = (s^1, \dots, s^N) \in \mathcal{S} = \mathcal{S}^1 \times \dots \times \mathcal{S}^N$ with $\mathcal{S}^i = \mathcal{S}^j \forall i, j \in \mathcal{N}$, where $s^i \in \mathcal{S}^i$ for any $i \in \mathcal{N}$.
- (ii) The state transition probability function is permutation invariant, i.e., $P(s_{t+1} | s_t, a_t) = P(Ms_{t+1} | Ms_t, Ma_t)$ for any $M \in \mathcal{M}$, where $s_t = (s_t^1, \dots, s_t^N)$, $s_{t+1} = (s_{t+1}^1, \dots, s_{t+1}^N)$, and $a_t = (a_t^1, \dots, a_t^N)$.
- (iii) The joint reward function is permutation preserving, i.e., $R(Ms_t, Ma_t) = MR(s_t, a_t)$ for any $M \in \mathcal{M}$, where $R(s, a) = (R^1(s, a), \dots, R^N(s, a))$.
- (iv) Each agent i obtains an observation function $o^i : \mathcal{S} \rightarrow \mathcal{O}$, which is bijective and has a common observation space $\mathcal{O}$ as its image space. The observations of the agents are permutation preserving, i.e., for any $s \in \mathcal{S}$ and $M \in \mathcal{M}$, $(o^1(Ms), \dots, o^N(Ms)) = M(o^1(s), \dots, o^N(s))$.

Remark 1 The decomposability of the global state serves as the basis of Definition 1, which is achievable due to the fact that the local state can be defined simply as the concatenation of an individual agent's local information and the common environmental information. The permutation preserving property of the rewards in condition (iii) implies that the average reward of the agents is permutation invariant. Moreover, the observation function defined in condition (iv) makes it possible for each agent to make different decisions under the same global state.

Denote $\pi(\cdot | s) = \prod_{i=1}^N \pi^i(\cdot | s)$ as the joint policy of the agents. Denote Π as the set of the joint policies. To avoid learning suboptimal near-deterministic policies, the entropy regularization technique is employed to encourage

exploration behaviors of agents. For a joint policy $\pi \in \Pi$, the entropy-regularized value function is defined as

$$V_\pi(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t (\bar{r}_{t+1} + \alpha \mathcal{H}(\pi(\cdot|s_t))) | s_0 = s \right], \quad (1)$$

where $a_t \sim \pi(\cdot|s_t)$, $s_{t+1} \sim P(\cdot|s_t, a_t)$, $\bar{r}_t = \frac{1}{N} \sum_{i \in \mathcal{N}} r_t^i$, $\alpha > 0$, and $\mathcal{H}(\pi(\cdot|s)) = -\sum_{a \in \mathcal{A}} \pi(a|s) \log(\pi(a|s))$ is the entropy functional [32]. Denote $\rho : \mathcal{S} \rightarrow [0, 1]$ as the initial state distribution of a trajectory. The objective of the agents is to collaboratively find a joint policy π through local information interaction such that the expectation of the value function (1) is maximized, which can be represented by

$$\max_{\pi \in \Pi} J(\pi) = (1 - \gamma) \mathbb{E}_{s \sim \rho} [V_\pi(s)]. \quad (2)$$

In addition, based on the value function (1), we can define an action-value function $Q_\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ w.r.t. policy π through the following Bellman equation [33]:

$$Q_\pi(s, a) = \bar{R}(s, a) + \gamma \mathbb{E}_{s' \sim P} [V_\pi(s')], \quad (3)$$

where $s' \sim P(\cdot|s, a)$ and $\bar{R}(s, a) = \mathbb{E}[\bar{r}_{t+1} | s_t = s, a_t = a]$. Based on formulas (1) and (3), it also holds that

$$V_\pi(s) = \mathbb{E}_{a \sim \pi} [Q_\pi(s, a) - \alpha \log(\pi(a|s))]. \quad (4)$$

Remark 2 Compared with the MARL objective functions in [23–26], the exploration capability of the policies is further considered in this paper through employing the entropy regularizer. The resulting optimization problem (2) is called an entropy-regularized MARL problem.

Remark 3 Due to the availability of only the local reward and policy information for each agent in N-MAMDPs, the global objective (2) should be solved in the distributed setting which requires each agent to improve its local policy through local communication.

4 Distributed Actor-Critic Algorithm Design

In this section, the property of policy optimality in homogeneous N-MAMDPs is studied, based on which a distributed actor-critic algorithm is proposed for the entropy-regularized MARL problem (2).

From condition (iv) of Definition 1, the local policy $\pi^i(\cdot|s)$ can be further designed as $\pi^i(\cdot|o^i(s))$ for any $s \in \mathcal{S}$ and $i \in \mathcal{N}$, where the agent first obtains its individual observation from the environment state, and then feeds the observation into the policy function to sample actions. Denote the observation-based policy as $\pi_o(\cdot|s) = \prod_{i=1}^N \pi^i(\cdot|o^i(s))$. Let $\Pi_o \subseteq \Pi$ be the set of the joint policies π_o . The following theorem states the property of policy optimality in homogeneous N-MAMDPs.

Theorem 1 In homogeneous N-MAMDPs, for the objective function defined in (2), it holds that

$$\max_{\pi \in \Pi} J(\pi) = \max_{\pi_o \in \Pi_o} J(\pi_o) = \max_{\pi_o \in \Pi_o : \pi^1 = \dots = \pi^N} J(\pi_o). \quad (5)$$

Proof. The first equation of (5) holds since the map o^i is bijective for any $i \in \mathcal{N}$ from Definition 1, that is, for the optimal joint policy $\pi_* \in \Pi$, there must exist a corresponding optimal policy $\pi_{o*} \in \Pi_o$, where $\pi_{o*}(\cdot|s) = \prod_{i=1}^N \pi_*^i(\cdot|o^i(s))$. For the second equation, it is obvious that for any permutation $M = [m_1, \dots, m_N] \in \mathcal{M}$ and $\pi_o(\cdot|s) \in \Pi_o$, if it satisfies that $\pi^i(\cdot|o^i(s)) = \pi^j(\cdot|o^j(Ms))$ with $i = m_j$ for any $s \in \mathcal{S}$ and $i \in \mathcal{N}$, then the policy entropy is permutation invariant, i.e., $\mathcal{H}(\pi_o(\cdot|s)) = \mathcal{H}(\pi_o(\cdot|Ms))$. Thus, due to the permutation invariance of the transition, the average reward and the policy entropy, there exists an optimal joint policy $\pi_{o*} \in \Pi_o$ such that $\pi_*^i(\cdot|o^i(s)) = \pi_*^j(\cdot|o^j(Ms))$. Furthermore, it holds that $o^i(s) = o^j(Ms) \in \mathcal{O}$ by condition (iv) of Definition 1. Thus, for any $i \in \mathcal{N}$ and $o \in \mathcal{O}$, it holds that $\pi_*^i(\cdot|o) = \pi_*^j(\cdot|o)$. Notice that this equation holds for arbitrary $M \in \mathcal{M}$, then we have $\pi_*^1(\cdot|o) = \dots = \pi_*^N(\cdot|o)$, which completes the proof. $\square$

Remark 4 The first equation of (5) implies that we can also consider the optimization problem (2) on the joint policy set Π_o . The second equation of (5) shows that for any $\pi_o \in \Pi_o$, suboptimality does not incur in the homogeneous N-MAMDP when the agents have the same local policy, which inspires us to incorporate a local policy sharing step into the distributed MARL algorithm design.

Based on Theorem 1, we now consider the optimization problem (2) on the policy set Π_o . Denote $\pi_\theta(\cdot|s) = \prod_{i=1}^N \pi_{\theta^i}^i(\cdot|o^i(s))$ as the parameterized joint policy with $\theta = [(\theta^1)^T, \dots, (\theta^N)^T]^T \in \Theta$, where $\theta^i \in \Theta^i$ and $\Theta = \prod_{i=1}^N \Theta^i$. Denote Π_Θ as the set that includes all π_θ . For notational convenience, we respectively denote the value function and the action-value function w.r.t. policy π_θ as $V_\theta(s)$ and $Q_\theta(s, a)$ for any $s \in \mathcal{S}$ and $a \in \mathcal{A}$. The objective function (2) can be equivalently rewritten as

$$\max_{\theta \in \Theta} J(\theta) = (1 - \gamma) \mathbb{E}_{s \sim \rho} [V_\theta(s)]. \quad (6)$$

Denote $\mathbb{P}_\theta(s_t)$ as the marginal state distribution w.r.t. policy π_θ . We first introduce the entropy-regularized policy gradient theorem [33].

Lemma 1 ([33]) Given any joint policy $\pi_\theta \in \Pi_\Theta$ and objective function $J(\theta)$ defined in (6), the gradient of $J(\theta)$ w.r.t. θ is represented by

$$\nabla_\theta J(\theta) = \mathbb{E}_{s \sim d_\theta, a \sim \pi_\theta} [\nabla_\theta \log(\pi_\theta(a|s)) [Q_\theta(s, a) - \alpha \log(\pi_\theta(a|s))]], \quad (7)$$

where $d_\theta(s) = (1-\gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}_\theta(s_t = s)$ is the visitation measure of state s .

Notice that $\theta = [(\theta^1)^T, \dots, (\theta^N)^T]^T$ and $\log(\pi_\theta(a|s)) = \sum_{i \in \mathcal{N}} \log(\pi_{\theta^i}^i(a^i|o^i(s)))$. The gradient of $J(\theta)$ w.r.t. θ can be further given as

$$\nabla_{\theta^i} J(\theta) = \mathbb{E}_{s \sim d_\theta, a \sim \pi_\theta} [[\nabla_{\theta^i} \log(\pi_{\theta^i}^i(a^i|o^i(s)))] [Q_\theta(s, a) - \alpha \log(\pi_\theta(a|s))]]. \quad (8)$$

Based on Theorem 1, let all the agents share the same structure of the local policy function, and it holds that $\Theta \subseteq \mathbb{R}^{Nm}$ with $\Theta^i \subseteq \mathbb{R}^m$ for any $i \in \mathcal{N}$. In addition, we define a copy operator $[\cdot] : \mathbb{R}^m \rightarrow \mathbb{R}^{Nm}$ such that for any $\hat{\theta} \in \mathbb{R}^m$, it holds that $[\hat{\theta}] = \mathbf{1} \otimes \hat{\theta}$, where $\mathbf{1} \in \mathbb{R}^N$.

We now develop a distributed actor-critic algorithm for the entropy-regularized MARL problem. Specifically, each agent i maintains a local critic $Q(s, a; \omega^i)$ to estimate the action-value function $Q_\theta(s, a)$, where $\omega^i \in \mathbb{R}^K$ is the policy parameter. Denote $C_{\omega, t} = [c_{\omega, t}(i, j)]_{N \times N}$ as the weight matrix for the critic. Once the on-policy transition $(s_t, a_t, \{r_{t+1}^i\}_{i \in \mathcal{N}}, s_{t+1}, a_{t+1})$ is sampled from the environment based on the policy π_θ , ω^i is updated using the TD error followed by a consensus update, which is represented by

$$\begin{cases} \tilde{\omega}_{t+1}^i = \omega_t^i + \beta_{\omega, t} \delta_t^i \nabla_{\omega} Q_t(\omega_t^i), \\ \omega_{t+1}^i = \sum_{j \in \mathcal{N}_t^i} c_{\omega, t}(i, j) \tilde{\omega}_{t+1}^j, \end{cases} \quad (9)$$

where $\beta_{\omega, t} > 0$ is the stepsize, $Q_t(\omega_t^i) = Q(s_t, a_t; \omega_t^i)$, and $\delta_t^i = r_{t+1}^i + \gamma(Q_{t+1}(\omega_t^i) - N\alpha \log(\pi_{\theta^i}^i(a_{t+1}^i|o^i(s_{t+1})))) - Q_t(\omega_t^i)$ is the local TD error. Let $C_{\theta, t} = [c_{\theta, t}(i, j)]_{N \times N}$ be the weight matrix for the actor. The actor parameter θ^i is updated inspired by the entropy-regularized multi-agent policy gradient formula (8) followed by a consensus update, which is expressed by

$$\begin{cases} \tilde{\theta}_{t+1}^i = \theta_t^i + \beta_{\theta, t} N \eta_t^i \psi_t^i, \\ \theta_{t+1}^i = \sum_{j \in \mathcal{N}_t^i} c_{\theta, t}(i, j) \tilde{\theta}_{t+1}^j, \end{cases} \quad (10)$$

where $\beta_{\theta, t} > 0$ is the stepsize, $\eta_t^i = Q_t(\omega_t^i) - \alpha \log(\pi_{\theta^i}^i(a_t|s_t))$, and $\psi_t^i = \nabla_{\theta^i} \log(\pi_{\theta^i}^i(a_t|o^i(s_t)))$.

Remark 5 Notice that the joint policy π_{θ_t} in (8) is replaced by $\pi_{[\theta_t]}$ in (10) since the joint parameter θ_t cannot be obtained locally. Furthermore, according to condition (iv) of Definition 1, observation $o^j(s_t)$ can be obtained by agent i through the observation $o^i(M_{s_t})$ with $m_i = j$ for any $i, j \in \mathcal{N}$, and it holds $\log(\pi_{[\theta_t]}(a_t|s_t)) = \sum_{j \in \mathcal{N}} \log(\pi_{\theta^j}^j(a_t^j|o^j(s_t)))$. Hence, each agent can update its actor in a distributed manner.

Remark 6 The local update of the critic (9) is designed to minimize the TD error, which is motivated by and has a similar form to that in [23]. However, due to the existence of the entropy regularizer, the calculation of the TD error in (9) is different from that in [23], and the convergence analysis should consider the corresponding modified Bellman equation. In addition, due to the homogeneity property of the agents, the policy consensus step is also considered in (10) which is inspired by [26]. However, [26] does not consider the entropy regularizer, and it is challenging to take the entropy of other agents' policies into consideration when updating the local policy parameter. Note that the entropy of the joint policy is estimated by each individual agent locally in (10).

5 Convergence Analysis

In this section, we establish the theoretical convergence results for the proposed algorithm (9), (10) based on the theory of stochastic approximation. Specifically, the convergence proof is provided considering the linear critic with the standard assumptions taken from [23, 26, 33].

Assumption 1 There exists $\kappa > 0$ such that $\pi_{\theta^i}^i(a^i|o^i(s)) \geq \kappa$ for any $s \in \mathcal{S}$, $a^i \in \mathcal{A}^i$ and $i \in \mathcal{N}$. $\pi_{\theta^i}^i(a^i|o^i(s))$ is also continuously differentiable w.r.t. θ^i . For any $\theta \in \Theta$, let P_θ be the state transition matrix of the Markov chain $\{s_t\}_{t \geq 0}$ w.r.t. policy π_θ , i.e., $P_\theta(s'|s) = \sum_{a \in \mathcal{A}} \pi_\theta(a|s) P(s'|s, a)$ $\forall s, s' \in \mathcal{S}$. It is assumed that the Markov chain $\{s_t\}_{t \geq 0}$ is irreducible and aperiodic under any policy π_θ .

Assumption 2 The weight matrix sequences $\{C_t\}_{t \geq 0}$ with $C_t \in \{C_{\omega, t}, C_{\theta, t}\}$ satisfy the following conditions: (i) C_t is row stochastic and $\mathbb{E}[C_t]$ is column stochastic for any $t \geq 0$. (ii) The spectral norm of $\mathbb{E}[C_t^T(I - \mathbf{1}\mathbf{1}^T/N)C_t]$ is strictly smaller than one. (iii) C_t is conditionally independent of r_{t+1}^i for any $i \in \mathcal{N}$ given the σ -algebra generated by the random variables before time t .

Assumption 3 The instantaneous reward r_{t+1}^i is uniformly bounded for any agent $i \in \mathcal{N}$ and $t \geq 0$.

Assumption 4 For the linear critic $Q(s, a; \omega) = \phi(s, a)^T \omega$, the feature vectors $\phi(s, a)$ are uniformly bounded for any $s \in \mathcal{S}$ and $a \in \mathcal{A}$. Furthermore, the feature matrix $\Phi \in \mathbb{R}^{N_s^a \times K}$ has full column rank, where $N_s^a = |\mathcal{S}| \times |\mathcal{A}|$.

Assumption 5 Stepsize sequences $\{\beta_{\omega, t}\}_{t \geq 0}$ and $\{\beta_{\theta, t}\}_{t \geq 0}$ satisfy $\beta_{\omega, t}, \beta_{\theta, t} > 0$, $\sum_t \beta_{\omega, t} = \infty$, $\sum_t \beta_{\theta, t} = \infty$, $\sum_t \beta_{\omega, t}^2 + \sum_t \beta_{\theta, t}^2 < \infty$, and $\beta_{\theta, t} = o(\beta_{\omega, t})$. In addition, it holds that $\lim_t \beta_{\omega, t+1} \beta_{\omega, t}^{-1} = \lim_t \beta_{\theta, t+1} \beta_{\theta, t}^{-1} = 1$.

Assumption 6 The critic update is stable almost surely (a.s.) for any $i \in \mathcal{N}$, i.e., $\sup_t \|\omega_t^i\| < \infty$ a.s. For the actor update, $\{\theta_t^i\}_{t \geq 0}$ belongs to a compact set including at least a local minimum of $J(\theta)$ for any $i \in \mathcal{N}$ and $t \geq 0$.

Remark 7 In Assumption 2, the weight matrices are only required to be column stochastic on average in condition (i). This allows us to employ the gossip type of communication schemes (e.g. the broadcast gossip scheme and the pairwise gossip scheme) for the multi-agent system. For these gossip schemes, condition (iii) holds if and only if the underlying communication topology is connected [34].

Remark 8 The stability assumption for the critic update can be further eliminated if the lower boundedness of the nonzero entries of $\{C_{\omega,t}\}_{t \geq 0}$ can be ensured, where the stability property of $\{\omega_t^i\}_{t \geq 0}$ can be theoretically obtained using the tools developed in [23] which are built upon the stochastic approximation theory.

Convergence of the critic. Based on these standard assumptions, the boundedness of the gradient of the actor parameter can be directly obtained [23], which allows us to use the two-time-scale stochastic approximation theory [35] to analyze the convergence of the faster update of ω_t^i with a fixed joint policy, i.e., $\pi_{\theta_t} \equiv \pi_\theta$.

Define $P_\theta \in \mathbb{R}^{N_s^a \times N_s^a}$ as the transition matrix of the state-action pairs with

$$P_\theta(s', a' | s, a) = P(s' | s, a) \pi_\theta(a' | s') \quad (11)$$

for any state-action pairs (s, a) and (s', a') . Let $\nu_\theta(s)$ be the stationary distribution of the states which satisfies $\nu_\theta(s) = \lim_t \mathbb{P}_\theta(s_t = s)$ for any $s \in \mathcal{S}$, then we can define a stationary distribution matrix $D_\theta^{s,a} = \text{diag}[\nu_\theta(s) \pi_\theta(a | s), s \in \mathcal{S}, a \in \mathcal{A}]$. Furthermore, let $\bar{R} = [\bar{R}(s, a), s \in \mathcal{S}, a \in \mathcal{A}] \in \mathbb{R}^{N_s^a}$ and $\Omega_\theta = [\alpha \log(\pi_\theta(a | s)), s \in \mathcal{S}, a \in \mathcal{A}] \in \mathbb{R}^{N_s^a}$. Then the Bellman operator $\mathcal{T}_\theta^Q : \mathbb{R}^{N_s^a} \rightarrow \mathbb{R}^{N_s^a}$ is defined for any action-value vector $Q = [Q(s, a), s \in \mathcal{S}, a \in \mathcal{A}]$ based on formulas (3) and (4), which is expressed by

$$\mathcal{T}_\theta^Q(Q) = \bar{R} + \gamma P_\theta(Q - \Omega_\theta). \quad (12)$$

Let $\mathcal{F}_{t,1} = \sigma(r_\tau, \omega_\tau, s_\tau, a_\tau, C_{\omega,\tau-1}, \tau \leq t)$, then $\{\mathcal{F}_{t,1}\}$ denotes a filtration, which is an increasing σ -algebra over time t . Denote $r_t = [r_t^1, \dots, r_t^N]^T$, $\omega_t = [(\omega_t^1)^T, \dots, (\omega_t^N)^T]^T$ and $\delta_t = [\delta_t^1, \dots, \delta_t^N]^T$. We can rewrite the update of ω_t in (9) in a compact form, which is expressed by

$$\omega_{t+1} = (C_{\omega,t} \otimes I)(\omega_t + \beta_{\omega,t} y_{t+1}), \quad (13)$$

where $I \in \mathbb{R}^{K \times K}$, $y_{t+1} = [\delta_t^1 \phi_t^T, \dots, \delta_t^N \phi_t^T]^T \in \mathbb{R}^{KN}$ and $\phi_t = \phi(s_t, a_t) \in \mathbb{R}^K$. For any positive integer L , we define an average operator as $\langle \cdot \rangle : \mathbb{R}^{LN} \rightarrow \mathbb{R}^L$ such that

$$\langle \omega \rangle = \frac{1}{N} (\mathbf{1}^T \otimes I) \omega = \frac{1}{N} \sum_{i \in \mathcal{N}} \omega^i, \quad (14)$$

where $\omega^i \in \mathbb{R}^L$ and $\omega = [(\omega^1)^T, \dots, (\omega^N)^T]^T \in \mathbb{R}^{LN}$. Let $J = \frac{1}{N} (\mathbf{1} \mathbf{1}^T \otimes I)$, then we have $J\omega_t = \mathbf{1} \otimes \langle \omega_t \rangle$, which denotes the agreement component in ω_t . Correspondingly, $\omega_{\perp,t} = \omega_t - \mathbf{1} \otimes \langle \omega_t \rangle$ denotes the disagreement component in ω_t .

Following the proof techniques in [23], we first establish the asymptotic consensus of ω_t^i for any $i \in \mathcal{N}$ through proving that $\omega_{\perp,t} \rightarrow 0$ a.s. as $t \rightarrow \infty$, which is shown in the following lemma.

Lemma 2 Under Assumptions 1-6, the asymptotic consensus of the critic is achieved almost surely, i.e., $\lim_t \omega_t^i - \langle \omega_t \rangle = 0$ a.s. for any $i \in \mathcal{N}$.

Proof. Based on the definition of $\omega_{\perp,t}$, we have $\omega_{\perp,t+1} = (I - J)\omega_{t+1} = (I - J)[(C_{\omega,t} \otimes I)(\mathbf{1} \otimes \langle \omega_t \rangle + \omega_{\perp,t} + \beta_{\omega,t} y_{t+1})]$. Since $C_{\omega,t}$ is row stochastic, it holds that $(I - J)(C_{\omega,t} \otimes I)(\mathbf{1} \otimes \langle \omega_t \rangle) = 0$, which leads to

$$\begin{aligned} \omega_{\perp,t+1} &= (I - J)[(C_{\omega,t} \otimes I)(\omega_{\perp,t} + \beta_{\omega,t} y_{t+1})] \\ &= [(I - \mathbf{1} \mathbf{1}^T / N) C_{\omega,t} \otimes I](\omega_{\perp,t} + \beta_{\omega,t} y_{t+1}). \end{aligned}$$

Let $M_{\omega,t} = C_{\omega,t}^T (I - \mathbf{1} \mathbf{1}^T / N)^T (I - \mathbf{1} \mathbf{1}^T / N) C_{\omega,t} \otimes I$ and ρ be the spectral norm of $\mathbb{E}[M_{\omega,t}]$, then it holds that $M_{\omega,t} = C_{\omega,t}^T (I - \mathbf{1} \mathbf{1}^T / N) C_{\omega,t} \otimes I$ and $\rho \in [0, 1)$ based on Assumption 2. Hence, we have

$$\begin{aligned} &\mathbb{E} [\|\beta_{\omega,t+1}^{-1} \omega_{\perp,t+1}\|^2 | \mathcal{F}_{t,1}] \\ &\leq \frac{\beta_{\omega,t}^2}{\beta_{\omega,t+1}^2} \mathbb{E} [(\beta_{\omega,t}^{-1} \omega_{\perp,t} + y_{t+1})^T M_{\omega,t} (\beta_{\omega,t}^{-1} \omega_{\perp,t} + y_{t+1}) | \mathcal{F}_{t,1}] \\ &\leq \frac{\beta_{\omega,t}^2}{\beta_{\omega,t+1}^2} \rho \mathbb{E} [(\beta_{\omega,t}^{-1} \omega_{\perp,t} + y_{t+1})^T (\beta_{\omega,t}^{-1} \omega_{\perp,t} + y_{t+1}) | \mathcal{F}_{t,1}], \end{aligned}$$

where the last inequality holds due to that $C_{\omega,t}$ and y_{t+1} are independent conditioning on $\mathcal{F}_{t,1}$ based on Assumption 2. By the Cauchy-Schwarz inequality, we further have that

$$\begin{aligned} &\mathbb{E} [\|\beta_{\omega,t+1}^{-1} \omega_{\perp,t+1}\|^2 | \mathcal{F}_{t,1}] \leq \frac{\beta_{\omega,t}^2}{\beta_{\omega,t+1}^2} \rho \|\beta_{\omega,t}^{-1} \omega_{\perp,t}\|^2 \\ &+ \frac{\beta_{\omega,t}^2}{\beta_{\omega,t+1}^2} \rho (2 \|\beta_{\omega,t}^{-1} \omega_{\perp,t}\| \mathbb{E} [\|y_{t+1}\| | \mathcal{F}_{t,1}] + \mathbb{E} [\|y_{t+1}\|^2 | \mathcal{F}_{t,1}]), \end{aligned}$$

where $\mathbb{E} [\|y_{t+1}\|^2 | \mathcal{F}_{t,1}] = \mathbb{E} [\sum_{i \in \mathcal{N}} \|\delta_t^i \phi_t\|^2 | \mathcal{F}_{t,1}]$ with $\delta_t^i = r_{t+1}^i + \gamma(\phi_{t+1}^T \omega_t^i - N \alpha \log(\pi_{\theta_t}^i(a_{t+1}^i | o^i(s_{t+1})))) - \phi_t^T \omega_t^i$. Due to the boundedness of the reward, the feature and the sampling probability of actions by the assumptions, we have that $\mathbb{E} [\|y_{t+1}\|^2 | \mathcal{F}_{t,1}]$ is bounded on the set $\{\sup_{\tau \leq t} \|\omega_\tau\| \leq M\}$ for any given $M > 0$ and $i \in \mathcal{N}$. Therefore, we can follow the proof of Lemma 5.3 in [23] to obtain that

$$\sup_t \mathbb{E} [\|\beta_{\omega,t}^{-1} \omega_{\perp,t}\|^2 \mathbb{I}_{\{\sup_t \|\omega_t\| \leq M\}}] < \infty,$$

based on which we can further obtain that $\lim_t \omega_{\perp,t} = 0$ a.s., i.e., $\lim_t \omega_t^i - \langle \omega_t \rangle = 0$ a.s. for any $i \in \mathcal{N}$. $\square$

Based on Lemma 2, we can provide the convergence result of the critic by analyzing the asymptotic behavior of the consensus vector $\langle \omega_t \rangle$, which is given in the following theorem.

Theorem 2 *Under Assumptions 1-6, for any policy π_θ and $i \in \mathcal{N}$, we have $\lim_t \omega_t^i = \omega_\theta$ a.s. with*

$$\omega_\theta = (\Phi^T D_\theta^{s,a} (I - \gamma P_\theta) \Phi)^{-1} (\Phi^T D_\theta^{s,a} (\bar{R} - \gamma P_\theta \Omega_\theta)). \quad (15)$$

Proof. Based on (13) and (14), we have

$$\begin{aligned} \langle \omega_{t+1} \rangle &= \langle (C_{\omega,t} \otimes I)(\omega_t + \beta_{\omega,t} y_{t+1}) \rangle \\ &= \langle \omega_t \rangle + \beta_{\omega,t} \langle (C_{\omega,t} \otimes I)(y_{t+1} + \beta_{\omega,t}^{-1} \omega_{\perp,t}) \rangle \\ &= \langle \omega_t \rangle + \beta_{\omega,t} \mathbb{E}[\langle \delta_t \rangle \phi_t | \mathcal{F}_{t,1}] + \beta_{\omega,t} \xi_{t+1}, \end{aligned} \quad (16)$$

where $\langle \delta_t \rangle = \bar{r}_{t+1} + \gamma (\phi_{t+1}^T \langle \omega_t \rangle - \alpha \log(\pi_{\theta_t}(a_{t+1}|s_{t+1}))) - \phi_t^T \langle \omega_t \rangle$, and $\xi_{t+1} = \langle (C_{\omega,t} \otimes I)(y_{t+1} + \beta_{\omega,t}^{-1} \omega_{\perp,t}) \rangle - \mathbb{E}[\langle \delta_t \rangle \phi_t | \mathcal{F}_{t,1}]$. To analyze the asymptotic behavior of $\langle \omega_t \rangle$ in (16) with the stochastic approximation theory, the assumptions in Appendix A are verified as follows. Firstly, assumption (1) is guaranteed since $\mathbb{E}[\langle \delta_t \rangle \phi_t | \mathcal{F}_{t,1}]$ is Lipschitz continuous in $\langle \omega_t \rangle$. Secondly, assumptions (2) and (3) are satisfied from Assumptions 1 and 5, respectively. In addition, since $\mathbb{E}[C_{\omega,t}]$ is column stochastic, it holds that $\mathbb{E}[\langle (C_{\omega,t} \otimes I) \omega_{\perp,t} \rangle | \mathcal{F}_{t,1}] = 0$. Thus, we have that

$$\begin{aligned} &\mathbb{E}[\langle (C_{\omega,t} \otimes I)(y_{t+1} + \beta_{\omega,t}^{-1} \omega_{\perp,t}) \rangle | \mathcal{F}_{t,1}] \\ &= \mathbb{E}[\langle (C_{\omega,t} \otimes I) y_{t+1} \rangle | \mathcal{F}_{t,1}] \\ &= \mathbb{E}[\langle y_{t+1} \rangle | \mathcal{F}_{t,1}] = \mathbb{E}[\langle \delta_t \rangle \phi_t | \mathcal{F}_{t,1}], \end{aligned} \quad (17)$$

i.e., $\{\xi_{t+1}\}_{t \geq 0}$ is a martingale difference sequence. Moreover, it holds that

$$\begin{aligned} \mathbb{E}[\|\xi_{t+1}\|^2 | \mathcal{F}_{t,1}] &\leq 2\mathbb{E}[\|y_{t+1} + \beta_{\omega,t}^{-1} \omega_{\perp,t}\|_{G_t}^2 | \mathcal{F}_{t,1}] \\ &\quad + 2\mathbb{E}[\|\langle \delta_t \rangle \phi_t | \mathcal{F}_{t,1}\|^2], \end{aligned}$$

where $G_t = \frac{1}{N^2} C_{\omega,t}^T \mathbf{1} \mathbf{1}^T C_{\omega,t} \otimes I$ and $\|\cdot\|_{G_t}$ is the Euclidean norm weighted by G_t . Recall that $\log(\pi_{\theta_t}(a_t|s_t))$ is bounded based on Assumption 1, we can follow similar steps in the proof of Theorem 4.6 in [23] to obtain that for any $M > 0$, it satisfies that

$$\mathbb{E}[\|\xi_{t+1}\|^2 | \mathcal{F}_{t,1}] \leq K(1 + \|\langle \omega_t \rangle\|^2)$$

for some $K < \infty$ over the set $\{\sup_t \|\omega_t\| \leq M\}$, which shows that assumption (4) is satisfied. Hence, the asymptotic behavior of $\langle \omega_t \rangle$ can be captured by the following

ODE:

$$\begin{aligned} \dot{\langle \omega \rangle} &= \Phi^T D_\theta^{s,a} (\bar{R} + \gamma P_\theta (\Phi \langle \omega \rangle - \Omega_\theta) - \Phi \langle \omega \rangle) \\ &= \Phi^T D_\theta^{s,a} (\gamma P_\theta - I) \Phi \langle \omega \rangle + \Phi^T D_\theta^{s,a} (\bar{R} - \gamma P_\theta \Omega_\theta). \end{aligned} \quad (18)$$

Note that all the eigenvalues of $(\gamma P_\theta - I)$ have negative real parts, and so does $\Phi^T D_\theta^{s,a} (\gamma P_\theta - I) \Phi$ since Φ has full column rank according to Assumption 4. Therefore, the ODE (18) is globally asymptotically stable with the equilibrium ω_θ defined in (15), which results in that $\lim_t \langle \omega_t \rangle = \omega_\theta$ a.s. Recall that $\lim_t \omega_t^i - \langle \omega_t \rangle = 0$ a.s. based on Lemma 2, then we have that $\lim_t \omega_t^i = \omega_\theta$ a.s. for any $i \in \mathcal{N}$. $\square$

It is worth noting that equation (15) can also be rewritten as $\Phi^T D_\theta^{s,a} (\mathcal{T}_\theta^Q(\Phi \omega_\theta) - \Phi \omega_\theta) = 0$, which shows that ω_θ is the minimizer of the Mean Square Projected Bellman Error (MSPBE) [36]:

$$\omega_\theta = \arg \min_{\omega} \|\Phi \omega - \Pi_\Phi \mathcal{T}_\theta^Q(\Phi \omega)\|_{D_\theta^{s,a}}^2, \quad (19)$$

where Π_Φ is an operator which projects any vector onto the space spanned by the columns of Φ , and $\|\cdot\|_{D_\theta^{s,a}}$ is the Euclidean norm weighted by $D_\theta^{s,a}$. Hence, Theorem 2 shows that each agent i is able to obtain a high-quality copy of the approximation of the global action-value function (3) using the distributed protocol (9).

Convergence of the actor. Following the two-time-scale stochastic approximation theory [35] and similar to the proof of Theorem 2 in [26], we can replace the critic parameter ω_t^i in (10) with the critic convergence point ω_{θ_t} when analyzing the convergence of the actors. Let $\eta_{t,\theta}^i = Q_t(\omega_\theta) - \alpha \log(\pi_{[\theta^i]}(a_t|s_t))$ and $\psi_{t,\theta}^i = \nabla_{\theta^i} \log(\pi_{\theta^i}(a_t^i|o^i(s_t)))$. Then we consider the actor update in the form of

$$\begin{cases} \tilde{\theta}_{t+1}^i = \theta_t^i + \beta_{\theta,t} N \eta_{t,\theta_t}^i \psi_{t,\theta_t}^i, \\ \theta_{t+1}^i = \sum_{j \in \mathcal{N}_t^i} c_{\theta,t}(i,j) \tilde{\theta}_{t+1}^j, \end{cases} \quad (20)$$

which can be rewritten as

$$\theta_{t+1} = (C_{\theta,t} \otimes I)(\theta_t + \beta_{\theta,t} y_{t,\theta_t}), \quad (21)$$

where $y_{t,\theta_t} = [N(\eta_{t,\theta_t}^1 \psi_{t,\theta_t}^1)^T, \dots, N(\eta_{t,\theta_t}^N \psi_{t,\theta_t}^N)^T]^T$. Similar to the proof of the critic convergence, θ_t is decomposed into $\theta_t = \mathbf{1} \otimes \langle \theta_t \rangle + \theta_{\perp,t}$ as well. We first show the asymptotic consensus of the actor parameters.

Lemma 3 *Under Assumptions 1-6, the asymptotic consensus of the actor is achieved almost surely, i.e., $\lim_t \theta_t^i - \langle \theta_t \rangle = 0$ a.s. for any $i \in \mathcal{N}$.*

Proof. For any $i \in \mathcal{N}$, η_{t,θ_t}^i is bounded since ω_{θ_t} is the MSPBE minimizer and $\pi_{\theta_t}^i(a_t^i|o^i(s_t)) \in [\kappa, 1)$ for any $a_t^i \in \mathcal{A}^i$ and $s_t \in \mathcal{S}$ according to Assumption 1. In addition, ψ_{t,θ_t}^i is also uniformly bounded since it is continuous over a compact set according to Assumptions 1 and 6. Hence, the boundedness of y_{t,θ_t} is guaranteed. Along the similar lines as the proof of Lemma 2, it can be obtained that $\lim_t \theta_{\perp,t} = 0$ a.s., which further leads to $\lim_t \theta_t^i - \langle \theta_t \rangle = 0$ a.s. for any $i \in \mathcal{N}$. $\square$

The consensus of the actor parameters allows us to establish the convergence of actor update by analyzing the asymptotic behavior of $\langle \theta_t \rangle$, which is shown in the following theorem.

Theorem 3 *Under Assumptions 1-6, for any $i \in \mathcal{N}$, the policy parameter θ_t^i converges almost surely to a point in the set of asymptotically stable equilibria of*

$$\dot{\hat{\theta}} = \mathbb{E}_{s_t \sim d_{[\hat{\theta}]}, a_t \sim \pi_{[\hat{\theta}]}} \left[\sum_{i \in \mathcal{N}} \eta_{t,[\hat{\theta}]}^i \psi_{t,[\hat{\theta}]}^i \right], \quad (22)$$

where $\hat{\theta} \in \mathbb{R}^m$.

Proof. Let $\mathcal{F}_{t,2} = \sigma(\theta_\tau, \tau \leq t)$ be the σ -field generated by $\{\theta_\tau, \tau \leq t\}$. Recall that $[\langle \theta_t \rangle] = \mathbf{1} \otimes \langle \theta_t \rangle$. In light of (16), we can rewrite (21) in the form of

$$\begin{aligned} \langle \theta_{t+1} \rangle &= \langle (C_{\theta,t} \otimes I)(\theta_t + \beta_{\theta,t} y_{t,\theta_t}) \rangle \\ &= \langle \theta_t \rangle + \beta_{\theta,t} \langle (C_{\theta,t} \otimes I)(y_{t,\theta_t} + \beta_{\theta,t}^{-1} \theta_{\perp,t}) \rangle \\ &= \langle \theta_t \rangle + \beta_{\theta,t} \mathbb{E}_{s_t \sim d_{[\langle \theta_t \rangle]}, a_t \sim \pi_{[\langle \theta_t \rangle]}} [\langle y_{t,[\langle \theta_t \rangle]} \rangle | \mathcal{F}_{t,2}] \\ &\quad + \beta_{\theta,t} \zeta_{t+1,1} + \beta_{\theta,t} \zeta_{t+1,2}, \end{aligned}$$

where $\zeta_{t+1,1}$ and $\zeta_{t+1,2}$ satisfy

$$\begin{cases} \zeta_{t+1,1} = \mathbb{E}_{s_t \sim d_{[\langle \theta_t \rangle]}, a_t \sim \pi_{[\langle \theta_t \rangle]}} [\langle y_{t,\theta_t} \rangle - \langle y_{t,[\langle \theta_t \rangle]} \rangle | \mathcal{F}_{t,2}], \\ \zeta_{t+1,2} = \langle (C_{\theta,t} \otimes I)(y_{t,\theta_t} + \beta_{\theta,t}^{-1} \theta_{\perp,t}) \rangle \\ \quad - \mathbb{E}_{s_t \sim d_{[\langle \theta_t \rangle]}, a_t \sim \pi_{[\langle \theta_t \rangle]}} [\langle y_{t,\theta_t} \rangle | \mathcal{F}_{t,2}]. \end{cases}$$

It can be directly obtained from Lemma 3 that $\zeta_{t+1,1} \rightarrow 0$ as $t \rightarrow \infty$. In addition, notice that $\{\zeta_{t+1,2}\}_{t \geq 0}$ is a martingale difference sequence due to that $\mathbb{E}[C_{\theta,t}]$ is column stochastic. Let $M_t = \sum_{\tau=0}^t \beta_{\theta,\tau} \zeta_{\tau+1,2}$, then $\{M_t\}_{t \geq 0}$ is a martingale sequence. Since both the sequences $\{\eta_{t,\theta_t}^i\}_{t \geq 0}$ and $\{\psi_{t,\theta_t}^i\}_{t \geq 0}$ are bounded, we can obtain that $\{\zeta_{t+1,2}\}_{t \geq 0}$ is bounded as well. Based on Assumption 5, we have $\sum_t \|M_{t+1} - M_t\|^2 = \sum_{t \geq 1} \|\beta_{\theta,t} \zeta_{t+1,2}\|^2 < \infty$. Along the similar lines as the proof of Theorem 4.7 in [23], we can apply the Kushner-Clark lemma and conclude that $\langle \theta_t \rangle$ converges a.s. to a point in the set of asymptotically stable equilibria of the ODE (22). Recall

that $\theta_t^i \rightarrow \langle \theta_t \rangle$ for any $i \in \mathcal{N}$, θ_t^i also converges a.s. to the point in the set of asymptotically stable equilibria of (22). $\square$

We further analyze the asymptotically stable equilibria of $\hat{\theta}$ in (22). Based on the definitions of $\eta_{t,\theta}^i$ and $\psi_{t,\theta}^i$, we have that

$$\begin{cases} \eta_{t,[\hat{\theta}]}^i = Q_t(\omega_{[\hat{\theta}]}) - \alpha \log(\pi_{[\hat{\theta}]}(a_t|s_t)), \\ \psi_{t,[\hat{\theta}]}^i = \nabla_{\hat{\theta}} \log(\pi_{[\hat{\theta}]}^i(a_t^i|o^i(s_t))). \end{cases} \quad (23)$$

Recall that $\log(\pi_{\theta}(a|s)) = \sum_{i \in \mathcal{N}} \log(\pi_{\theta}^i(a^i|o^i(s)))$, the ODE (22) can be further rewritten as

$$\begin{aligned} \dot{\hat{\theta}} &= \mathbb{E}_{s_t \sim d_{[\hat{\theta}]}, a_t \sim \pi_{[\hat{\theta}]}} [[\nabla_{\hat{\theta}} \log(\pi_{[\hat{\theta}]}(a_t|s_t))] \\ &\quad [Q_t(\omega_{[\hat{\theta}]}) - \alpha \log(\pi_{[\hat{\theta}]}(a_t|s_t))]]. \end{aligned} \quad (24)$$

On the other hand, we directly consider the joint policy $\pi_{[\hat{\theta}]}$ with all agents sharing the same policy function as well as the same parameter $\hat{\theta} \in \mathbb{R}^m$ based on Theorem 1. In this case, we can treat (6) as an optimization problem w.r.t. $\hat{\theta}$. Based on the entropy-regularized policy gradient (7), the gradient of $J([\hat{\theta}])$ w.r.t. parameter $\hat{\theta}$ can be determined by

$$\begin{aligned} \nabla_{\hat{\theta}} J([\hat{\theta}]) &= \mathbb{E}_{s \sim d_{[\hat{\theta}]}, a \sim \pi_{[\hat{\theta}]}} [[\nabla_{\hat{\theta}} \log(\pi_{[\hat{\theta}]}(a|s))] \\ &\quad [Q_{[\hat{\theta}]}(s, a) - \alpha \log(\pi_{[\hat{\theta}]}(a|s))]]. \end{aligned} \quad (25)$$

Therefore, $\hat{\theta}$ will converge to a local optimum of $J([\hat{\theta}])$ if it evolves based on the ODE $\dot{\hat{\theta}} = \nabla_{\hat{\theta}} J([\hat{\theta}])$. Based on (24) and (25), ODE (22) can be finally rewritten as

$$\dot{\hat{\theta}} = \nabla_{\hat{\theta}} J([\hat{\theta}]) + \mathbb{E}_{s \sim d_{[\hat{\theta}]}, a \sim \pi_{[\hat{\theta}]}} [\nabla_{\hat{\theta}} \log(\pi_{[\hat{\theta}]}(a|s)) v_{[\hat{\theta}]}(s, a)], \quad (26)$$

where $v_{[\hat{\theta}]}(s, a) = Q(s, a; \omega_{[\hat{\theta}]}) - Q_{[\hat{\theta}]}(s, a)$ is the non-zero approximation error term. Since $v_{[\hat{\theta}]}(s, a)$ is small by Theorem 2, it can be concluded that $\hat{\theta}$ in (22) will converge to a small neighborhood of a local optimum of $J([\hat{\theta}])$, which is the best result one can hope for any single-agent actor-critic algorithm using linear function approximation [23]. To empirically verify the convergence conclusion, several experiments are conducted on a toy environment, which is shown in Appendix B. By comparing our distributed actor-critic algorithm with a centralized actor-critic algorithm developed from (25), it is confirmed that our distributed algorithm has close learning performance as the centralized algorithm with convergence guarantee.

6 Practical Algorithm Design

In this section, a practical distributed actor-critic algorithm is proposed based on the theoretical distributed entropy-regularized algorithm in Section 4 considering scalability, data efficiency and learning stability.

Notice that the learning performance of the theoretical distributed algorithm can be severely limited by the standard assumptions in Section 5, such as the linear function approximator, the on-policy training architecture, and even the decaying learning rate. Hence, we combine this algorithm with the famous DRL training architecture [10] to develop a practical distributed algorithm, where the modifications are shown as follows:

Actor and critic networks. To improve the learning performance, we design both the actor and critic as neural networks (NNs). Let $o_t^i = o^i(s_t)$. Then each agent i samples actions from the actor NN, i.e., $a_t^i \sim \pi_{\theta^i}(\cdot|o_t^i)$. We take the joint observation and action of all agents as the input of the critic NN to alleviate the non-stationarity issue in MARL [10], such that the critic NN can be represented by $Q_{\omega^i}(\{o_t^k, a_t^k\}_{k \in \mathcal{N}})$. Nevertheless, it makes the critic hard to scale when simply concatenating the agents' observations and actions in a user-specified order as the input. In order to improve the scalability of the algorithm, the graph convolutional network (GCN) [37] is utilized to design the critic NN. In the GCN, each agent is regarded as a vertex containing local information in a graph, and each edge represents an information interaction channel between a pair of agents. The l -th graph convolutional layer is expressed by

$$h_l = \sigma \left(h_{l-1} W_{s,l} + \frac{1}{N} A_{\text{adj}} h_{l-1} W_{o,l} \right), \quad (27)$$

where $h_l \in \mathbb{R}^{N \times K_l}$ and $h_{l-1} \in \mathbb{R}^{N \times K_{l-1}}$ denote the output and input of this layer, respectively, $A_{\text{adj}} \in \mathbb{R}^{N \times N}$ denotes the adjacency matrix of the graph, $W_{s,l}, W_{o,l} \in \mathbb{R}^{K_{l-1} \times K_l}$ are the learnable graph filters, and σ is a point-wise nonlinear function. Considering the complex interaction among the agents, A_{adj} is a matrix of all ones with zero diagonal elements, which means that the graph is complete. Moreover, in the first graph convolutional layer, the i -th row of input h_0 is the concatenation of o_t^i and a_t^i . For the critic NN of agent i , L graph convolutional layers are utilized to extract the information from the graph, where $h_{i,L} = [(h_{i,L}^1)^T, \dots, (h_{i,L}^N)^T]^T \in \mathbb{R}^{N \times K_L}$ denotes the output of the L -th layer. In light of the value decomposition technique [9], a linear layer f_{v^i} is then employed for mapping each $h_{i,L}^k$ to the estimated local scalar critic value Q^k . Denote $o_t = (o_t^1, \dots, o_t^N)$. Thus, we can represent the GCN-based critic NN of agent i as $Q_{\omega^i}(o_t, a_t) = \sum_{k \in \mathcal{N}} f_{v^i}(h_{i,L}^k)$, where ω^i is the concatenation of $\{v^i\}$ and the GCN filter tensors $\{W_{s,l}\}_l, \{W_{o,l}\}_l$. Notice that the proposed critic NN is scalable

since the size of the learnable parameter ω^i is irrelevant to the number of agents. In addition, [it can be found that the joint observation is used in the critic NN instead of the environment state](#). This is due to that important local information could be omitted in the environment-provided state, and the loss of essential local information could severely degrade the learning performance [38].

Local parameter update. The distributed algorithm proposed in Section 4 could be data-inefficient due to its on-policy nature. To effectively leverage the sampled experience, we develop an off-policy version of this algorithm based on the DRL architecture. Denote $\mathcal{B}^i = \{(o_t, a_t, r_{t+1}^i, o_{t+1}^i)\}_t$ and $Q_{\bar{\omega}^i}(o_t, a_t)$ as the replay buffer and the target critic NN of agent i , respectively. Based on formula (8), the local update of ω^i is represented by the following loss function:

$$J_Q(\omega^i) = \mathbb{E}_{(o_t, a_t, r_{t+1}^i, o_{t+1}^i) \sim \mathcal{B}^i} [(Q_{\omega^i}(o_t, a_t) - y^i)^2], \quad (28)$$

where y^i is the one-step TD target, and it is represented by $y^i = r_{t+1}^i + \gamma(Q_{\bar{\omega}^i}(o_{t+1}, a_{t+1}) - N \alpha \log(\pi_{\theta^i}(a_{t+1}^i | o_{t+1}^i)))$ with $a_{t+1}^j \sim \pi_{\theta^j}(\cdot | o_{t+1}^j) \forall j \in \mathcal{N}$. Notice that y_i can be calculated in a distributed manner since the local policy π_{θ^i} is used to generate the actions of all agents at timestep $t+1$, which makes sense due to that all the actor NNs will achieve consensus by Theorem 3. Denote $\hat{\nabla}_{\omega^i} J_Q(\omega^i)$ as the stochastic gradient of (28). As for the actor NN, instead of updating θ^i based on (10) directly, we consider a loss function whose gradient w.r.t. θ^i has the same form as (8) under the DRL architecture, and then minimize this loss function with the reparameterization trick for improving learning stability [39]. Specifically, the actor loss function has the form of

$$J_{\pi}(\theta^i) = \mathbb{E}_{o_t \sim \mathcal{B}^i, a_t \sim \pi_{\theta^i}} [\alpha \log(\pi_{\theta^i}(a_t | o_t)) - Q_{\omega^i}(o_t, a_t)], \quad (29)$$

whose relation with (8) is explicitly analyzed in Appendix C. Next, each local policy π_{θ^i} is reparameterized using a NN transformation $a_t^i = f_{\theta^i}(o_t^i, \varepsilon_t^i)$, where ε_t^i is an input noise vector sampled from some predefined distributions. Denote $\mathcal{X}$ as the distribution of the joint noise $\varepsilon_t = (\varepsilon_t^1, \dots, \varepsilon_t^N)$. Denote $f_{\theta}(o_t, \varepsilon_t) = (f_{\theta^1}(o_t^1, \varepsilon_t^1), \dots, f_{\theta^N}(o_t^N, \varepsilon_t^N))$. Therefore, the loss function (29) can be rewritten as

$$J_{\pi}(\theta^i) = \mathbb{E}_{o_t \sim \mathcal{B}^i, \varepsilon_t \sim \mathcal{X}} [\alpha \log(\pi_{\theta^i}(f_{\theta}(o_t, \varepsilon_t) | o_t))] - \mathbb{E}_{o_t \sim \mathcal{B}^i, \varepsilon_t \sim \mathcal{X}} [Q_{\omega^i}(o_t, f_{\theta}(o_t, \varepsilon_t))]. \quad (30)$$

Also, due to the consensus of the local policies, the gradient of (30) w.r.t. θ^i can also be calculated in a distributed fashion by assuming $\pi_{\theta^j}^j = \pi_{\theta^i}^i$ for any $j \in \mathcal{N} \setminus \{i\}$. Denote $\hat{\nabla}_{\theta^i} J_{\pi}(\theta^i)$ as the stochastic gradient of (30). In practice, we further multiply $\hat{\nabla}_{\theta^i} J_{\pi}(\theta^i)$ by N to match up

Algorithm 1 DER-AC

```

1: Initialize  $\{\omega^i\}_{i=1}^N, \{\theta^i\}_{i=1}^N$ 
2: Set  $\mathcal{B}^i \leftarrow \emptyset$  and  $\tilde{\omega}^i \leftarrow \omega^i$  for each  $i \in \mathcal{N}$ 
3: for episode  $m = 1, 2, \dots, L$  do
4:   for  $t = 0, \dots, T - 1$  do
5:     Sample  $a_t^i \sim \pi_{\theta^i}(\cdot | o_t^i)$  for each  $i \in \mathcal{N}$ 
6:     Update state  $s_{t+1} \sim P(\cdot | s_t, a_t)$ 
7:     Obtain  $o_{t+1}$  and  $r_{t+1}^i$  for each  $i \in \mathcal{N}$ 
8:      $\mathcal{B}^i \leftarrow \mathcal{B}^i \cup \{(o_t, a_t, r_{t+1}^i, o_{t+1})\}$  for each  $i \in \mathcal{N}$ 
9:   end for
10:  for  $i = 1, 2, \dots, N$  do
11:    Sample a batch of data  $\mathcal{D}^i$  from  $\mathcal{B}^i$ 
12:     $\omega^i \leftarrow \omega^i - \lambda_Q \hat{\nabla}_{\omega^i} J_Q(\omega^i)$ 
13:     $\theta^i \leftarrow \theta^i - \lambda_\pi N \hat{\nabla}_{\theta^i} J_\pi(\theta^i)$ 
14:     $\alpha^i \leftarrow \alpha^i - \lambda_\alpha \hat{\nabla}_{\alpha^i} J(\alpha^i)$ 
15:     $\tilde{\omega}^i \leftarrow \tau \omega^i + (1 - \tau) \tilde{\omega}^i$ 
16:  end for
17:  for  $i = 1, 2, \dots, N$  do
18:     $\tilde{\omega}^i \leftarrow \sum_{j \in \mathcal{N}_m^i} c_{\omega, m}(i, j) \omega^j$ 
19:     $\tilde{\theta}^i \leftarrow \sum_{j \in \mathcal{N}_m^i} c_{\theta, m}(i, j) \theta^j$ 
20:  end for
21:  Set  $\omega^i \leftarrow \tilde{\omega}^i$  and  $\theta^i \leftarrow \tilde{\theta}^i$  for each  $i \in \mathcal{N}$ 
22: end for
23: Return  $\{\omega_*^i\}_{i=1}^N, \{\theta_*^i\}_{i=1}^N$ 

```

with formula (10). In addition, the mechanism of automatic entropy adjustment is incorporated in our algorithm for balancing exploration and exploitation during the training process [40]. Denote α^i as the local temperature of agent i . We adjust α^i by minimizing the following loss function:

$$J(\alpha^i) = \mathbb{E}_{o_t^i \sim \mathcal{B}^i, a_t^i \sim \pi_{\theta^i}} \left[-\alpha^i (\log(\pi_{\theta^i}(a_t^i | o_t^i)) + \mathcal{H}_0) \right], \quad (31)$$

where $\mathcal{H}_0 > 0$ denotes the target entropy. Denote $\hat{\nabla}_{\alpha^i} J(\alpha^i)$ as the stochastic gradient of (31).

Other training settings. Following the basic settings in existing DRL algorithms [39, 41], the parameters of the target critic NN $\tilde{\omega}^i$ are updated by calculating the exponentially moving average of ω^i , and the update stepsizes of parameters are set as constants. In addition, both the gradient-based and the consensus-based parameter updates will be performed after each episode concludes. Our algorithm distributed entropy-regularized actor-critic (DER-AC) is presented in Algorithm 1 with the overall structural diagram illustrated in Fig. 1. Compared with the algorithm proposed in Section 4, the NN-based approximator and the DRL-based training architecture enable the application of DER-AC to complex MARL tasks with continuous state and action spaces.

Remark 9 To enhance sample efficiency and learning stability, DRL-based algorithms commonly incorporate

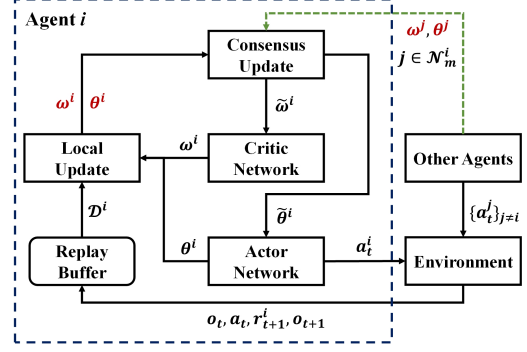


Fig. 1. The overall structural diagram of DER-AC. The green dashed arrow represents the information sharing process by communication, and the notations in red denote the NN parameters after the local update step.

the experience replay technique [10, 11], which requires additional memory to store past experiences. In our practical algorithm, the memory complexity of each agent is $\mathcal{O}(H(2P + Q + 1))$, where H denotes the upper bound on the number of the stored experiences, P and Q denote the dimensions of the joint observation and the joint action, respectively.

7 Experiments

In this section, the MARL tasks considered in this paper are introduced at first. Then, the details of the experimental setup, results, ablations, and the adaptation under the local observable setting are presented.

Environments. We evaluate the proposed algorithm DER-AC on three standard multi-robot coordination tasks [42, 43], which are coverage control, formation control and triangle control (see Fig. 2). These tasks are constructed in the multi-agent particle environment (MPE) [10, 44], where the agents follow discrete second-order dynamics and move in a continuous 2D space with collision awareness. Each task has two versions with 6 agents and 15 agents, respectively, and the source code has been released for reproducibility¹. We describe these tasks briefly below:

- **Coverage Control:** There are N agents and N landmarks in the environment. The agents should cover the landmarks as quickly as possible while avoiding collisions.
- **Formation Control:** There are N agents and one landmark in the environment. The agents should distribute themselves evenly on a circumference centered at one landmark as quickly as possible while avoiding collisions.
- **Triangle Control:** There are N agents and three landmarks in the environment, where the landmarks determine the position of an equilateral triangle. Each

¹ <https://github.com/KrisHYF/Envs-for-DER-AC>

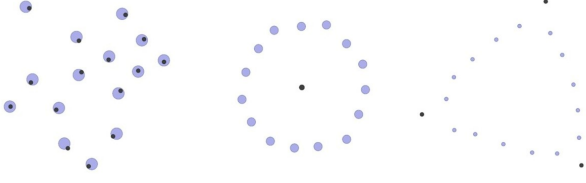


Fig. 2. Three coordination tasks constructed in the MPE environment, where each purple disk denotes an agent and each black disk denotes a landmark. Left: the coverage control task. Middle: the formation control task. Right: the triangle control task.

agent should safely position itself in one side of the triangle as quickly as possible such that the agents in this side can spread out equally between two corresponding landmarks.

In all the three tasks, the observation of an agent includes its absolute position and velocity, the relative positions of the other agents as well as the relative positions of the landmarks, such that these MPE environments satisfy the definition of homogeneous N-MAMDP due to the full observability and the permutation preserving property of the observations [26]. It is worth pointing out that the imperfect state information is used as the observations in the tasks with 15 agents since the policy can still generate high quality coordination behaviors with local observation information after training [26, 44]. In addition, the continuous action spaces are considered in all the tasks, which requires higher exploration ability for algorithms to find the optimal policy parameter. Realize that these tasks are indeed planning problems, and no explicit state transition information is provided during the coordinating process in MPE. Therefore, neither control-based methods nor model predictive control-based planning methods are suitable to solve these tasks.

Experimental setup. In the experiments, two state-of-the-art MARL algorithms PIC [44] and MASAC [45] are used as the baselines, where both algorithms follow the paradigm of CTDE. Specifically, PIC uses the policy sharing strategy, such that the policy parameter can be trained using a single-agent RL algorithm. In our experiments, the training algorithm DDPG [41] in PIC is replaced by SAC [40] for improving its learning performance. MASAC is a variant of MADDPG [10], which considers each agent as an independent learner equipped with SAC instead of DDPG. For the distributed algorithms, the DRL-based algorithm distributed deterministic actor-critic (D2-AC) [26] is implemented, which [shares a similar training architecture with ours](#), but locally updates the policy based on a deterministic policy gradient approach. In addition, a variant of DER-AC is also utilized as a baseline, in which the entropy regularizer is not incorporated (i.e. $\alpha = 0$) and is thus named by distributed actor-critic (D-AC). [Notice that these three distributed algorithms have the same memory complexity.](#) The differences between DER-AC and

the baselines are summarized in Table 1. In our implementation, all the algorithms share the same structures of both the critic and actor NNs. The critic NN follows the structure in Section 6 with the hyperparameters $L = 2$ and $K_1 = K_2 = 128$, and the double Q-learning trick [46] is employed to further improve the learning stability by reducing the bias of the value estimation. The actor NN is modeled as a Gaussian policy which consists of two hidden layers and two linear layers², where each hidden layer contains 128 neurons with ReLU as the activation function. In the actor NN, the observation is first processed by the hidden layers and then fed into the linear layers to calculate the mean and the standard deviation of the action, respectively. For the distributed algorithms, we generate the communication network $\mathcal{G}_t = (\mathcal{N}, \mathcal{E}_t)$ through placing communication links among the agents randomly [23], where the number of links $|\mathcal{E}_t|$ can be determined based on formula $|\mathcal{E}_t| = \frac{1}{2}\rho N(N-1)$ with a pre-defined connectivity ratio $\rho \in [0, 1]$. For weight matrices $C_t \in \{C_{\omega,t}, C_{\theta,t}\}$, the element $c_t(i, j)$ is calculated by

$$c_t(i, j) = \begin{cases} \{1 + \max[d_t(i), d_t(j)]\}^{-1}, & (i, j) \in \mathcal{E}_t \\ 1 - \sum_{j \in \mathcal{N}_t^i} c_t(i, j), & i \in \mathcal{N}, \end{cases} \quad (32)$$

where $d_t(i) = |\mathcal{N}_t^i|$ denotes the degree of agent i . Note that each agent i can calculate $c_t(i, j)$ in (32) for any $j \in \mathcal{N}$ in a distributed manner. We set $\rho = 0.2, 0.4$ for the tasks with 6 agents and 15 agents, respectively. The other hyperparameters of the distributed algorithms are tuned to be optimal, which are shown in Appendix D. Notice that the hyperparameters of PIC and MASAC are also finely tuned. As a unique hyperparameter of DER-AC, the initial value of α plays an important role in determining the final performance in our experiments. A relatively larger α could always result in a better policy after training, which could be attributed to the agents' improved exploration ability. However, a substantial decrease in the learning pace is observed when setting α to excessively high values. In addition, we set smaller learning rates for the algorithms when the number of agents increases, which results in the better learning stability. Finally, all the algorithms are implemented with PyTorch 1.11.0, which are both trained and tested on a laptop equipped with the Windows 10 operating system, an 11th generation Intel(R) Core(TM) i7-11800H CPU, and an NVIDIA GeForce RTX 3060 GPU.

Results. We evaluate the learning performance of each algorithm by conducting five independent trials on each version of the tasks, where the smoothed learning curves are shown in Fig. 3. It is clearly verified in the results that the learning performance of the distributed algorithm can be significantly improved by incorporating the entropy regularizer, as DER-AC outperforms D-AC

² <https://github.com/pranz24/pytorch-soft-actor-critic>

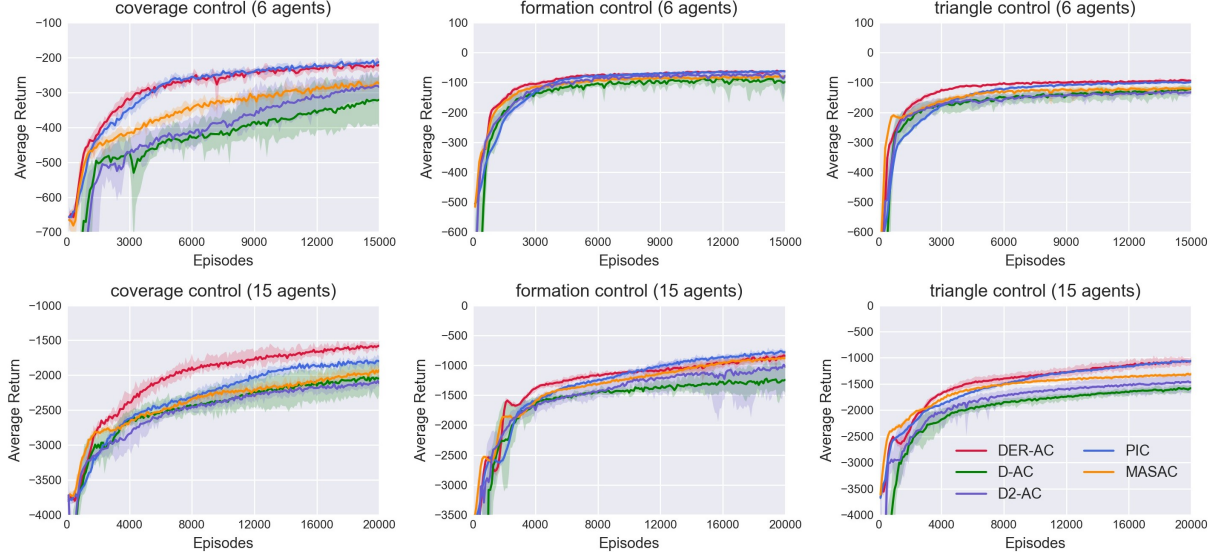


Fig. 3. Learning curves of DER-AC, D-AC, D2-AC, PIC and MASAC in three tasks, where the solid line represents the average return obtained from five independent training runs, while the shaded regions correspond to the upper and lower bounds of the returns achieved during the training process.

Table 1
Comparison of DER-AC with four baselines

	Entropy	Distributed	Number of Critics	Number of Actors
DER-AC (ours)	✓	✓	N	N
D-AC (ours)		✓	N	N
D2-AC [26]		✓	N	N
PIC [44]	✓		1	1
MASAC [45]	✓		N	N

in both the average return and the learning stability in all the cases, where the poor learning stability of D-AC could be due to the insufficient exploration during the training process, which results in that the learned policy will be easily trapped in the suboptimal policies. On the contrary, DER-AC is able to bypass these suboptimal policies, which contributes to the better sample efficiency due to that all the agents act as randomly as possible. Note that in most of the cases, the learning capability of D2-AC falls between that of DER-AC and D-AC. Nevertheless, it faces the similar exploration issues as D-AC since the exploring policies are near deterministic. Both PIC and MASAC have the better learning stability than the distributed algorithms, which could be due to the centralized training architecture. Both DER-AC and PIC outperform MASAC by significant margins in most cases, which could be explained by that achieving policy consensus could accelerate the policy learning in MARL environments with homogeneous agents. Both DER-AC and MASAC learn faster than PIC at the early stage of the training process in most cases. This phenomenon can be explained by that the agents equipped with different policies can sample richer experiences, which is beneficial for initial policy learning. It is worth pointing out that DER-AC outperforms PIC in the coverage control

task with 15 agents, which does not conflict with Theorem 1 since the observation function is not bijective in this case and DER-AC has the potential to train better policies based on the richer experiences. We further compare these algorithms in the view of communication requirements, where the notations ϑ and ℓ are respectively used to quantify the communication costs of transmitting local actor and critic information once. In DER-AC, each agent has $\rho(N-1)$ neighbors in expectation, such that its expected communication cost at each step is $\rho(N-1)(\vartheta + \ell)$. In PIC, a centralized unit is utilized to transmit the policy to all the agents, such that the communication cost of this unit at each step is $N\vartheta$. Note that the centralized unit requires additional communication resources to receive observations, actions, and rewards from all the agents. In MASAC, each agent should send its local policy to all the other agents for the local critic update, such that the communication graph is fully connected, and the communication cost of each agent is therefore $(N-1)\vartheta$. Compared with the CTDE-based algorithms, DER-AC can handle the large communication overheads for large-scale MARL tasks by setting a small ρ , which can also tolerate the communication link failures during the training process. As a consequence, DER-AC is the most desirable due to both the excellent learning performance and the mild communication requirements.

Ablations. In our ablation study, we explore the influence of communication traffic on both the distributed algorithms DER-AC and D-AC. Specifically, we run each algorithm on the three environments with 15 agents under the connectivity ratios $\rho = 0, 1$, respectively. Notice that the agents will not perform the parameter consen-

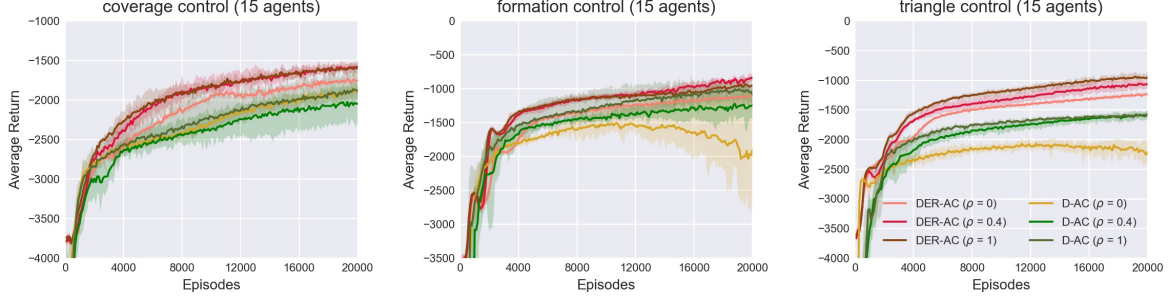


Fig. 4. Comparison of learning performance of DER-AC and D-AC under different connectivity ratios.

sus update for $\rho = 0$. On the contrary, they will perform all-to-all communication for parameter consensus for $\rho = 1$, which requires the most communication resources during the training process. The learning curves of both DER-AC and D-AC under different ρ are represented in Fig. 4. It is demonstrated in this figure that it is important to perform the parameter consensus in DER-AC, as DER-AC with $\rho = 0$ has the worst learning performance. Furthermore, the learning performance of DER-AC may not be enhanced by increasing the connectivity ratio blindly, since the algorithm with $\rho = 1$ does not outperform that with $\rho = 0.4$ in both coverage control and formation control, which inspires us to set a small connectivity ratio ρ at the beginning when deploying DER-AC in new environments. D-AC requires the parameter consensus step as well, since its learning performance could decrease in both formation control and triangle control when $\rho = 0$. It is worth pointing out that DER-AC outperforms D-AC in all the environments given the same connectivity ratio. This result further verifies the importance of incorporating the entropy regularizer into the training process.

Local observable setting. DER-AC can be adapted under the local observable setting, where each agent only has access to its own observation o_t^i and action a_t^i . Hence, the critic NN is reconstructed as an MLP which takes (o_t^i, a_t^i) as its input. Denote DER-AC-L as the adapted version of DER-AC under this setting. To evaluate DER-AC-L, the coverage control task with 6 agents is modified under the local observable setting, where each agent here can only observe two closest agents and three closest landmarks, such that the observation function o^i is not bijective. It can be found in Fig. 5 that DER-AC-L performs badly when no communication exists among the agents, which indicates that the local observable setting poses a huge challenge for policy learning. However, DER-AC-L shows close learning performance to PIC, DER-AC and D2-AC when the parameter consensus mechanism can be incorporated, where the last three algorithms require more information, i.e., the joint observation-action information $\{(o_t^i, a_t^i)\}_{i \in \mathcal{N}}$, for training. It is worth pointing out that DER-AC-L outperforms D2-AC at the early stage of the training process, which shows excellent sample efficiency due to the

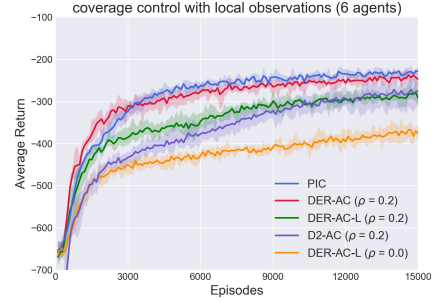


Fig. 5. Learning curves of PIC, DER-AC, DER-AC-L and D2-AC in the coverage control task under the local observation setting.

entropy regularization. The simulation results demonstrate the potential of employing our approach to solve practical MARL tasks under both local observable and decentralized settings.

8 Conclusion and Discussion

In this paper, an entropy-regularized MARL problem has been investigated, where policy sharing is shown to provably preserve the optimality in homogeneous N-MAMDPs. An on-policy distributed actor-critic algorithm with local policy consensus has been proposed which enjoys the asymptotic convergence guarantee with linear function approximation. For practical usage, a distributed off-policy version of the proposed algorithm has been developed based on the DRL training architecture which considers scalability, data efficiency and learning stability. The empirical results demonstrate that the proposed distributed algorithm has the comparable learning performance to several centralized-trained algorithms under limited communication resources.

Even though our practical algorithm shows the potential to deal with MARL tasks under the local observable setting, the non-stationarity issue cannot be neglected as the individual critic NN only takes the local information as its input. Therefore, our future work is to propose distributed MARL algorithms which can effectively

eliminate the non-stationarity issue under the local observable setting.

Appendix

A Stochastic Approximation Theory

Consider the stochastic approximation iteration

$$x_{t+1} = x_t + \gamma_t[h(x_t, Y_t) + M_{t+1} + \beta_{t+1}], \quad t \geq 0, \quad (\text{B.1})$$

where $x_t \in \mathbb{R}^n$, the sequence $\{\gamma_t\}_{t \geq 0}$ satisfies $\gamma_t > 0$, and $\{Y_t\}_{t \geq 0}$ is a Markov chain defined on a finite set A . Consider the following assumptions:

- (1) Function $h : \mathbb{R}^n \times A \rightarrow \mathbb{R}^n$ is Lipschitz continuous in its first argument;
- (2) Markov chain $\{Y_t\}_{t \geq 0}$ is irreducible with stationary distribution π ;
- (3) The stepsize sequence $\{\gamma_t\}_{t \geq 0}$ satisfies $\sum_t \gamma_t = \infty$ and $\sum_t \gamma_t^2 < \infty$;
- (4) $\{M_t\}_{t \geq 0}$ is a martingale difference sequence satisfying $\mathbb{E}[\|M_{t+1}\|^2 | x_t, Y_t, \tau \leq t] \leq K(1 + \|x_t\|^2)$ for some $K > 0$ and $t \geq 0$;
- (5) The sequence $\{\beta_t\}_{t \geq 0}$ is a bounded random sequence with $\lim_t \beta_t = 0$ almost surely.

Consider the ODE

$$\dot{x} = \bar{h}(x) = \sum_{i \in A} \pi(i) h(x, i), \quad (\text{B.2})$$

where x^* is the unique globally asymptotically stable equilibrium. We introduce the following lemmas [35].

Lemma 4 Under assumptions (1)-(5), suppose that

$$\lim_{c \rightarrow \infty} \frac{\bar{h}(cx)}{c} = h_\infty(x) \quad (\text{B.3})$$

exists on compact sets uniformly for some $h_\infty \in C(\mathbb{R}^n)$. If the origin is the unique globally asymptotically stable equilibrium of the ODE $\dot{y} = h_\infty(y)$, then it holds that

$$\sup_t \|x_t\| < \infty \text{ a.s.} \quad (\text{B.4})$$

Lemma 5 Under assumptions (1)-(5), if $\sup_t \|x_t\| < \infty$ a.s., then $x_t \rightarrow x^*$ a.s.

B Experiments on a Toy Environment

The environment. The environment contains an even number of N agents, where each agent i obtains a fixed initial local state $s^i = \cos(\frac{i-1}{N-1}\pi)$ with a common action space $\{0, 1\}$. After the first timestep, the global state

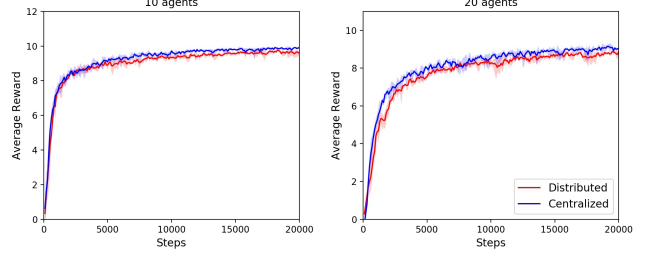


Fig. B.1. Learning curves of both the distributed and centralized algorithms in the toy environment with 10 agents and 20 agents, respectively.

$s = (s^1, \dots, s^N)$ will transition to a terminal state. All the agents share the same reward function, which is expressed by

$$R^i(s, a) = 10 \left[\text{mean}_{\{i: i \leq \frac{N}{2}\}} \mathbb{I}_{\{a^i=1\}} - \text{mean}_{\{i: i > \frac{N}{2}\}} \mathbb{I}_{\{a^i=1\}} \right].$$

It is apparent that the agents can get the largest reward 10 when $a^i = 1$ for $i \leq N/2$ and $a^i = 0$ for $i > N/2$. In addition, for each agent $i \in \mathcal{N}$, the observation is designed as $o^i = [s^i, \cos(\frac{0}{N-1}\pi), \dots, \cos(\frac{N-1}{N-1}\pi)]$. Note that this environment can be cast as a homogeneous N-MAMDP [26].

Experimental setup. We compare our distributed algorithm (9) and (10) with a centralized algorithm, which follows the single-agent actor-critic training architecture developed based on (25). Based on [26], the feature function of the linear critic is designed as $\phi(s, a) = \text{concat}[\{s^i, \text{one_hot}(a^i)\}_i]$, and the actor is considered as a Boltzman policy [23], where the softmax function is utilized to generate the distribution over two actions. Both algorithms are evaluated on this environment with $N = 10, 20$, respectively. In order to achieve effective learning, we update the parameters using the Adam optimizer with adaptive learning rates. In addition, we set $\alpha = 0.01$ and construct both the weight matrices $C_{\omega, t}$ and $C_{\theta, t}$ following the similar steps in Section 7 with $\rho = 0.4$.

Results. The learning curves of both algorithms are shown in Fig. B.1. Both algorithms are capable of learning a policy that is close to the optimal policy in a stable manner. Furthermore, our distributed algorithm has the similar learning performance as the centralized algorithm, which verifies the convergence conclusion in Section 5 empirically.

C Gradient of the Actor Loss Function

Denote $G_\theta(o_t, a_t) = \alpha \log(\pi_\theta(a_t|o_t)) - Q_{\omega^i}(o_t, a_t)$. Then, the gradient of (29) w.r.t. θ^i is represented by $\nabla_{\theta^i} J_\pi(\theta^i) = \mathbb{E}_{o_t \sim \mathcal{B}^i} [\sum_{a_t} \nabla_{\theta^i} (\pi_\theta(a_t|o_t) G_\theta(o_t, a_t))]$.

Table D.1

Hyperparameter Settings of the Distributed Algorithms

Hyperparameters	6 Agents	15 Agents
Number of training episodes L	15000	20000
Episode length T	25	25
Learning rate λ_Q	0.01	0.005
Learning rate λ_π	0.005	0.001
Learning rate λ_α	0.005	0.001
Discount factor γ	0.95	0.95
Target smoothing coefficient τ	0.01	0.01
Initial value of α^i	1	1
Target entropy $\mathcal{H}_0$	-2	-2
Buffer size $ \mathcal{B}^i $	375000	500000
Batch size $ \mathcal{D}^i $	256	256

Note that

$$\begin{aligned}
& \sum_{a_t} \nabla_{\theta^i} (\pi_{\theta}(a_t|o_t) G_{\theta}(o_t, a_t)) \\
&= \sum_{a_t} (\nabla_{\theta^i} \pi_{\theta}(a_t|o_t)) G_{\theta}(o_t, a_t) + \pi_{\theta}(a_t|o_t) \nabla_{\theta^i} G_{\theta}(o_t, a_t) \\
&= \sum_{a_t} \pi_{\theta}(a_t|o_t) (\nabla_{\theta^i} \log(\pi_{\theta^i}(a_t^i|o_t^i))) (G_{\theta}(o_t, a_t) + \alpha) \\
&= \sum_{a_t} \pi_{\theta}(a_t|o_t) (\nabla_{\theta^i} \log(\pi_{\theta^i}(a_t^i|o_t^i))) G_{\theta}(o_t, a_t),
\end{aligned}$$

where the last equation holds based on formula (3.6) in [23]. Thus, the gradient of (29) w.r.t. θ^i can be finally expressed by

$$\nabla_{\theta^i} J_{\pi}(\theta^i) = \mathbb{E}_{o_t \sim \mathcal{B}^i, a_t \sim \pi_{\theta}} [[\nabla_{\theta^i} \log(\pi_{\theta^i}(a_t^i|o_t^i))] [\alpha \log(\pi_{\theta}(a_t|o_t)) - Q_{\omega^i}(o_t, a_t)]].$$

It can be found that $-\nabla_{\theta^i} J_{\pi}(\theta^i)$ differs from gradient (8) in the distribution of observations. However, it is a common trick to sample states or observations from the memory to calculate gradients in the state-of-the-art DRL algorithms [10, 41], such that (29) is used to update the policy parameters in the practical algorithm of this paper.

D Hyperparameter Settings

See Table D.1.

References

- [1] Pang, B., Jiang, Z. P., & Mareels, I. (2020). Reinforcement learning for adaptive optimal control of continuous-time linear periodic systems. *Automatica*, 118, 109035.
- [2] Haarnoja, T., Ha, S., Zhou, A., Tan, J., Tucker, G., et al. (2019). Learning to walk via deep reinforcement learning. In *Robotics: Science and Systems*.
- [3] Wu, C., Pan, W., Staa, R., Liu, J., Sun, G., et al. (2023). Deep reinforcement learning control approach to mitigating actuator attacks. *Automatica*, 152, 110999.
- [4] Zhang, K., Yang, Z., & Basar, T. (2019). Multi-agent reinforcement learning: A selective overview of theories and algorithms. *arXiv preprint arXiv:1911.10635*.
- [5] Ren, W., & Beard, R. W. (2005). Consensus seeking in multiagent systems under dynamically changing interaction topologies. *IEEE Transactions on Automatic Control*, 50(5), 655–661.
- [6] Wen, G., Yu, X., Yu, W., & Lü, J. (2021). Coordination and control of complex network systems with switching topologies: A survey. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 51(10), 6342–6357.
- [7] Zhou, Y., Wen, G., Wan, Y., & Fu, J. (2023). Consensus tracking control for a class of general linear hybrid multi-agent systems: A model-free approach. *Automatica*, 156, 111198.
- [8] Wen, G., Lv, Y., Zheng, W. X., Zhou, J., & Fu, J. (2023). Joint robustness of time-varying networks and its applications to resilient consensus. *IEEE Transactions on Automatic Control*, doi: 10.1109/TAC.2023.3237493.
- [9] Sunehag, P., Lever, G., Gruslys, A., Czarnecki, W. M., Zambaldi, V., et al. (2018). Value-decomposition networks for cooperative multiagent learning based on team reward. In *International Conference on Autonomous Agents and MultiAgent Systems*, 2085–2087.
- [10] Lowe, R., Wu, Y., Tamar, A., Harb, J., Abbeel, P., et al. (2017). Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in Neural Information Processing Systems*, 6382–6393.
- [11] Iqbal, S., & Sha, F. (2019). Actor-attention-critic for multi-agent reinforcement learning. In *International Conference on Machine Learning*, 2961–2970.
- [12] Gupta, J. K., Egorov, M., & Kochenderfer, M. (2017). Cooperative multi-agent control using deep reinforcement learning. In *International Conference on Autonomous Agents and MultiAgent Systems*, 66–83.
- [13] Khan, A., Kumar, V., & Ribeiro, A. (2021). Large scale distributed collaborative unlabeled motion planning with graph policy gradients. *IEEE Robotics and Automation Letters*, 6(3), 5340–5347.
- [14] Stanković, M. S., Beko, M., & Stanković, S. S. (2023). Distributed consensus-based multi-agent temporal-difference learning. *Automatica*, 151, 110922.
- [15] Sha, X., Zhang, J., You, K., Zhang, K., & Başar, T. (2022). Fully asynchronous policy evaluation in distributed reinforcement learning over networks. *Automatica*, 136, 110092.
- [16] Pennesi, P., & Paschalidis, I. C. (2010). A distributed actor-critic algorithm and applications to mobile sensor network coordination problems. *IEEE Transactions on Automatic Control*, 55(2): 492–497.
- [17] Stanković, M. S., Beko, M., & Stanković, S. S. (2022). Convergent distributed actor-critic algorithm based on gradient temporal difference. In *European Signal Processing Conference*, 2066–2070.
- [18] Liu, W., Niu, H., Jang, I., Herrmann, G., & Carrasco, J. (2022). Distributed neural networks training for robotic manipulation with consensus algorithm. *IEEE Transactions on Neural Networks and Learning Systems*, doi: 10.1109/TNNLS.2022.3191021.
- [19] Macua, S. V., Tukiainen, A., Hernández, D. G.-O., Baldazo, D., de Cote, E. M., et al. (2020). Diff-DAC: Distributed actor-critic for average multitask deep reinforcement learning. *arXiv preprint arXiv:1710.10363*.

- [20] Stanković, M. S., Beko, M., Ilić, N., & Stanković, S. S. (2022). Multi-agent actor-critic multitask reinforcement learning based on GTD(1) with consensus. In *IEEE Conference on Decision and Control*, 4591–4596.
- [21] Zeng, S., Doan, T. T., & Romberg, J. (2023). Finite-time convergence rates of decentralized stochastic approximation with applications in multi-agent and multi-task learning. *IEEE Transactions on Automatic Control*, 68(5): 2758–2773.
- [22] Zhang, Y., & Zavlanos, M. M. (2019). Distributed off-policy actor-critic reinforcement learning with policy consensus. In *IEEE Conference on Decision and Control*, 4674–4679.
- [23] Zhang, K., Yang, Z., Liu, H., Zhang, T., & Basar, T. (2018). Fully decentralized multi-agent reinforcement learning with networked agents. In *International Conference on Machine Learning*, 5872–5881.
- [24] Suttle, W., Yang, Z., Zhang, K., Wang, Z., Başar, T., et al. (2020). A multi-agent off-policy actor-critic algorithm for distributed reinforcement learning. *IFAC-PapersOnLine*, 53(2), 1549–1554.
- [25] Grosnit, A., Cai, D., & Wynter, L. (2021). Decentralized deterministic multi-agent reinforcement learning. In *IEEE Conference on Decision and Control*, 1548–1553.
- [26] Chen, D., Li, Y., & Zhang, Q., Communication-efficient actor-critic methods for homogeneous markov games, In *International Conference on Learning Representations*, 2022.
- [27] Yang, H., Zhao, J., Lam, K.-Y., Xiong, Z., Wu, Q., et al. (2022). Distributed deep reinforcement learning based spectrum and power allocation for heterogeneous networks. *IEEE Transactions on Wireless Communications*, 21(9), 6935–6948.
- [28] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.
- [29] Hu, Y., Fu, J., & Wen, G. (2023). Safe reinforcement learning for model-reference trajectory tracking of uncertain autonomous vehicles with model-based acceleration. *IEEE Transactions on Intelligent Vehicles*, 8(3), 2332–2344.
- [30] Zhang, T., Li, Y., Wang, C., Xie, G., & Lu, Z. (2021). Fop: Factorizing optimal joint policy of maximum-entropy multi-agent reinforcement learning. In *International Conference on Machine Learning*, 12491–12500.
- [31] Hu, Y., Fu, J., & Wen, G. (2023). Graph soft actor-critic reinforcement learning for large-scale distributed multirobot coordination. *IEEE Transactions on Neural Networks and Learning Systems*, doi: 10.1109/TNNLS.2023.3329530.
- [32] Geist, M., Scherrer, B., & Pietquin, O. (2019). A theory of regularized markov decision processes. In *International Conference on Machine Learning*, 2160–2169.
- [33] Suttle, W., Yang, Z., Zhang, K., & Liu, J. (2019). Stochastic convergence results for regularized actor-critic methods. *arXiv preprint arXiv:1907.06138*.
- [34] Bianchi, P., & Jakubowicz, J. (2013). Convergence of a multi-agent projected stochastic gradient algorithm for non-convex optimization. *IEEE Transactions on Automatic Control*, 58(2), 391–405.
- [35] Borkar, V. S. (2009). *Stochastic approximation: A dynamical systems viewpoint*. Hindustan Publ. Agency and Cambridge Uni. Press.
- [36] Dann, C., Neumann, G., & Peters, J. (2014). Policy evaluation with temporal differences: A survey and comparison. *Journal of Machine Learning Research*, 15, 809–883.
- [37] Gama, F., Isufi, E., Leus, G., & Ribeiro, A. (2020). Graphs, convolutions, and neural networks: From graph filters to graph neural networks. *IEEE Signal Processing Magazine*, 37(6), 128–138.
- [38] Yu, C., Velu, A., Vinitisky, E., Wang, Y., Bayen, A., et al. (2022). The surprising effectiveness of ppo in cooperative multi-agent games. In *Advances in Neural Information Processing Systems*, 24611–24624.
- [39] Haarnoja, T., Zhou, A., Abbeel, P., & Levine, S. (2018). Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International Conference on Machine Learning*, 1861–1870.
- [40] Haarnoja, T., Zhou, A., Hartikainen, K., Tucker, G., Ha, S., et al. (2018). Soft actor-critic algorithms and applications. *arXiv preprint arXiv:1812.05905*.
- [41] Lillicrap, T. P., Hunt, J. J., Pritzel, A., Heess, N., Erez, T., et al. (2016). Continuous control with deep reinforcement learning. In *International Conference on Learning Representations*, 1–14.
- [42] Balch, T., & Arkin, R. C. (1998). Behavior-based formation control for multirobot teams. *IEEE Transactions on Robotics and Automation*, 14(6), 926–939.
- [43] Agarwal, A., Kumar, S., Sycara, K., & Lewis, M. (2020). Learning transferable cooperative behavior in multi-agent teams. In *International Conference on Autonomous Agents and MultiAgent Systems*, 1741–1743.
- [44] Liu, I. J., Yeh, R. A., & Schwing, A. G. (2020). PIC: permutation invariant critic for multi-agent deep reinforcement learning. In *Conference on Robot Learning*, 590–602.
- [45] Willemsen, D., Coppola, M., & de Croon, G. C. (2021). MAMBPO: Sample-efficient multi-robot reinforcement learning using learned world models. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 5635–5640.
- [46] Hasselt, H. V., Guez, A., & Silver, D. (2016). Deep reinforcement learning with double Q-Learning. In *AAAI Conference on Artificial Intelligence*, 2094–2100.